

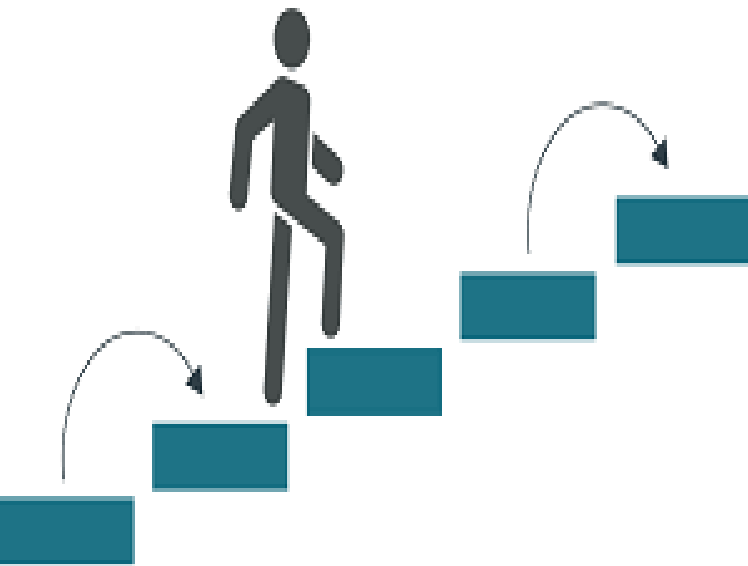


Good Morning!

formations



Devops Transformation Workshop/Bootcamp



By Syed Azhar (Devops Evangelist)
Founder: tformations
Ex. Director Devops Careem

Course Overview

- This twenty (24) hours course provides an introduction to DevOps – the cultural and professional movement that stresses communication, collaboration and integration between Business analyst, Project Manager/Scrum master , software developers, Testers and IT operations professionals. This course is designed to provide the basic education / awareness needed to transforms an individual or organization to DevOps culture and to understand its principles, techniques, practices and approach. Improved workflows will result in an improved ability to design, develop, deploy and operate software and services faster.
- It will help you understand and adopt devops transformation which can enable a tech organization to become more competitive by reducing **mean time to deliver values without compromising the Quality.**

What will you learn:

- Best practices of Agile which leads towards devops culture
- Devops Foundation and Vocabulary.
- Benefits to the business
- Concepts and practices — including its relationship to Agile, Lean and IT Service Management (ITSM)
- 5 Cs of Devops and how you can leverage those techniques and practices within organization.
- Execute CICD pipelines using Jenkins/ TFS/ Azure and other deployment tools.
- Applying and scaling DevOps in an enterprise environment
- Improved workflows, communication and feedback loop.
- Collaboration Model “Where Work Happen” for ultimate DevOps culture
- Optimize operations by leveraging virtualization, containers, and cloud technologies – Azure/AWS
- Reliance on Automation- Self Service
- Critical success factors and key performance indicators-



Material

- DevOps – The Basics (pre-class resource)
- Learner Manual (excellent post-class reference)
- Participation in unique exercises designed to apply concepts
- Sample documents, templates, tools and techniques
- Access to additional sources of information and communities

Who should attend:

- Work experience as an DevOps Engineer, Ops Engineer, Developer, PM, Scrum Master DevOps Manager, Tester, Delivery Manager, Hands-On Test Manager, or other roles that ensure the development cycle on software projects is achieved through optimized schedules, costs, and quality.

Training Mode:

This training include Interactive Session, open discussion, Real Case Study Labs, Handson Session , Demos , MVP, Presentations.

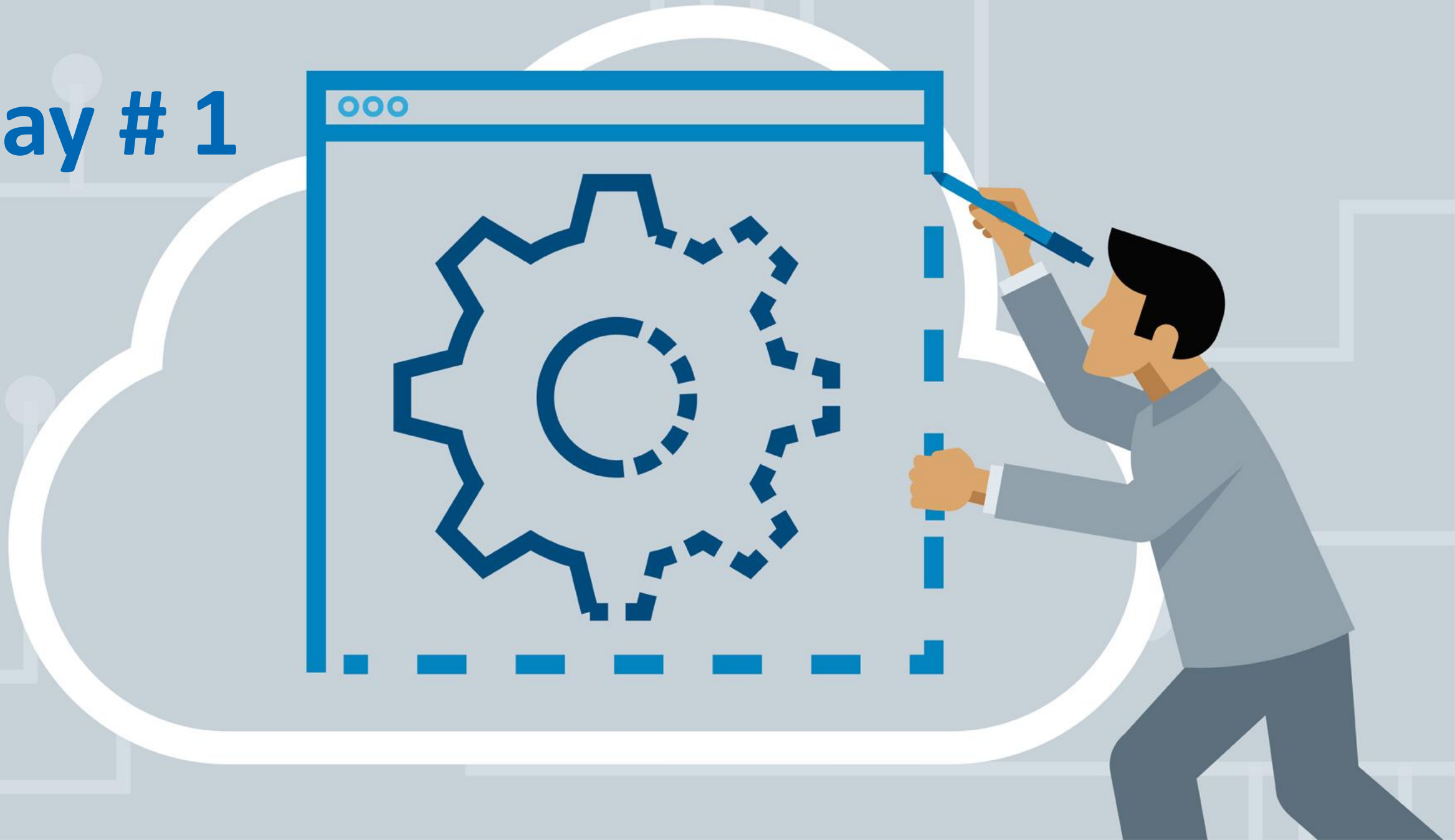
Note: *Few topics required hands-on labs might not relevant for PM or Scrum masters, Managers hence parallel sessions will be provided as per role which addresses the business cases study of Devops and how to implement devops practices within team.*

Course Outline

1. Agile “Practitioner approach”
2. Devops Foundations
3. Devops Practices (5 Cs of Devops) and adoptions
4. CI/CD Pipeline (Delivery Pipeline)
5. Devops-Toolchaining
6. Collaboration model-Working together
7. Cloud Adoption- AWS and Azure workshops
8. Self Service and Automation
9. Operations (Monitoring & Alerting, APM)
10. DevSecOps



Day # 1



Agile “Practitioner approach”

- **Objective**

To train people on Agile methodology and Scrum process, make them enable for Agile, tell them the real philosophy of agile and scrum, so everyone will be on same page so we can get maximum benefit of this framework.

- To provide in depth understanding and thought process of Scrum artifacts, Scrum Roles, Scrum ceremonies , Sprints , Story points so one can relate that to daily activities.

- **Why Agile?**

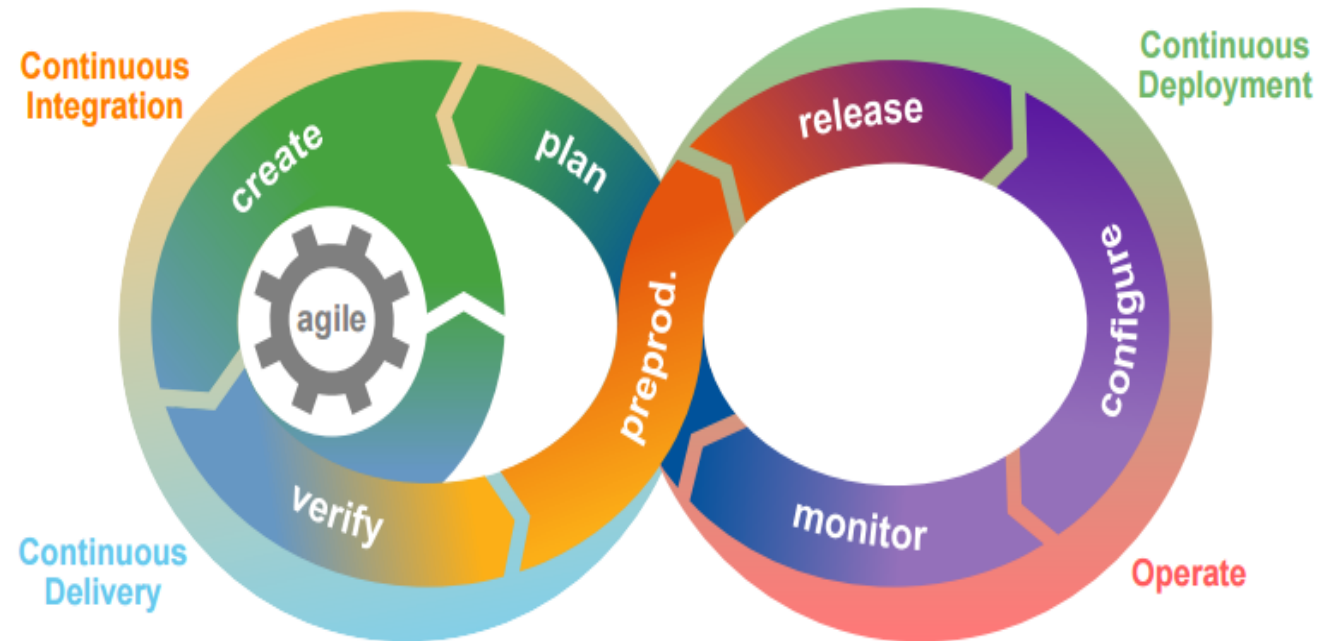
- Background
- Agile Manifesto
- Agile Principles
- Practical coaching and workshop

- **Introduction to Scrum**

- Roles
- Artifacts
- Ceremonies
- Practical Lab sessions with practitioners

- **Estimation**

- Story Points
- Real Case study – Proven Methodology



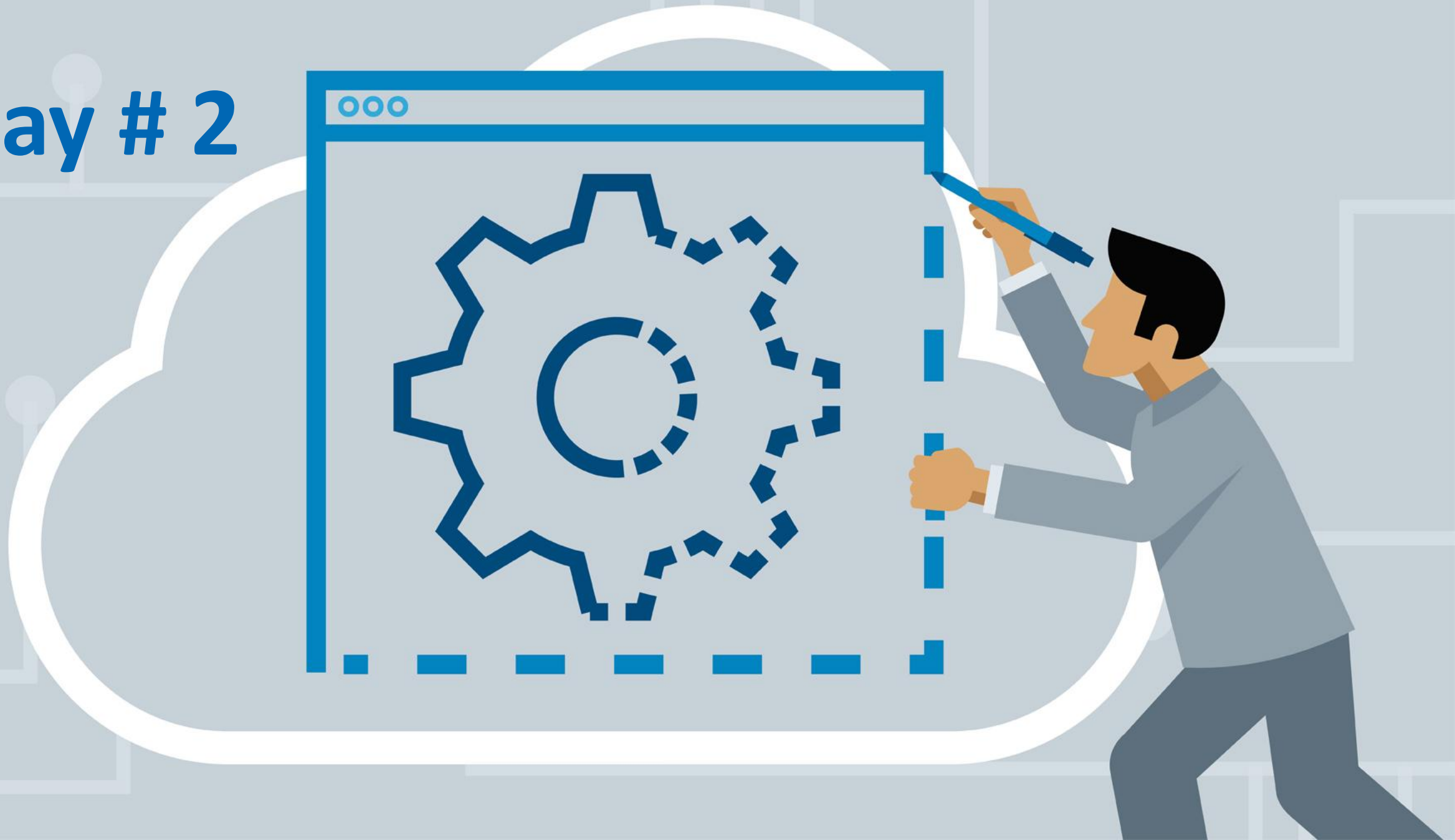
Devops Foundation

- SDLC
- Non Continuous Delivery Model
- Background – Why Devops
- What is Devops
- DevOps objectives and vocabulary
- Why it is essential for Tech World – Challenges
- Assessment and adoptions of Devops Culture
- Practical Exercise
- Devops 2018 Study Report

Duration : 4 hours
Content & Material :
Presentation and
Workshop



Day # 2



Devops Practices: 5 Cs of Devops

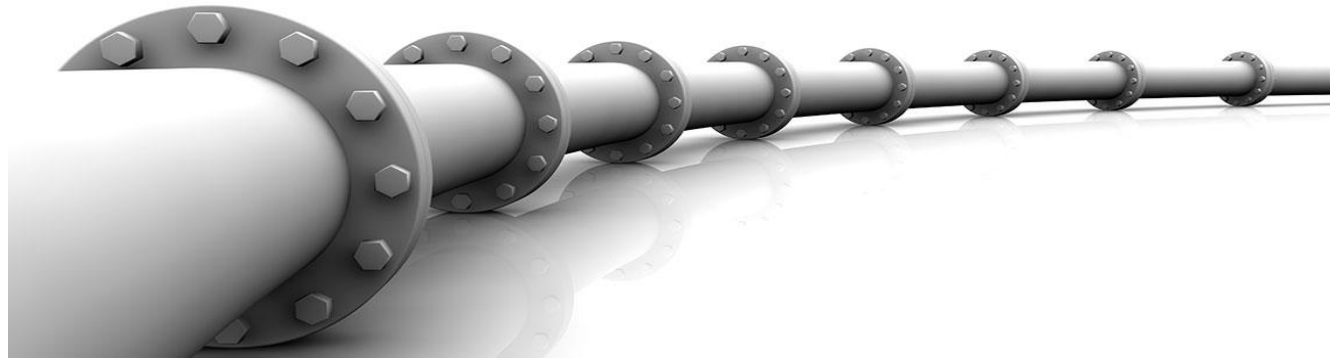
- Continuous Planning
 - MS Project, TFS , JIRA, Integrations , Collaboration and Case Study
- Continuous Integration (Development)
 - TFS, SVN, Github, Phabricator, Builds Target, Release Management, Artifactory Management, CI Process (Jenkins, Teamcity, TFS) and Unit Testing (Sonarcubes)
- Continuous Testing
 - Gated Code Promotion Process: Automation of AAC, Regression Testing , QA Environment Provisioning, Staging Env, Testing Strategy by Martin Fowler (Contract Testing and E2E Testing) : Appium , Selenium webdriver
- Continuous Delivery
 - Infrastructure as Code, Continuous Deployment, Configuration Automation, Blue- Green Deployment, Canary Release, Auto Rollback , Tools: Terraform, Puppet, Salt, Cloudformations using AWS , Dockerization Tools: Kubernetes
- Continuous Feedback & Monitoring
 - Production Deployment automation, 24 7 Techops automation, Application performance management, Real time monitoring and alerting, Auto Recovery, Self healing, Reliability and Scalability, Alerting tools, Incident Management Process
- Implementing 5 Cs “ Real Case Study”
 - Demo, Practices and Report to be present for the organization who has successfully implemented above 5Cs : Techniques, tooling, Process, Benefits



CI/CD Pipeline (Continuous Delivery)

- **Hands-on Workshop: (Using common tools) Parallel Session for non Tech user on Devops business case**

- Plumbing Techniques -Implementation Approach and consideration
- Leveraging CD practices
- Configuration Management
- Delivery pipeline using Jenkins
- Plumbing Hurdles
- Configuration
- Multiple Case Studies
- Deployment Techniques



- **Followed by Practical Implementation:**
- On the Basis of above learning and concepts conduct a real time implementation
 1. Form multiple teams (atleast 4 to 5 members)
 2. Balance the team with as per skill set and Role(Dev, SM, PM, TechOps, Test Engineers)
 3. Take a real case from existing projects or product team has.
 4. Design a delivery pipeline
 5. Develop an MVP with the help of tools and trainers supervision
 6. Demo , learning and Q& A

Duration : 2 hours
Content & Material :
Presentation and
Real Case Study : Lab
and Assignment

Devops Culture- Collaboration Model

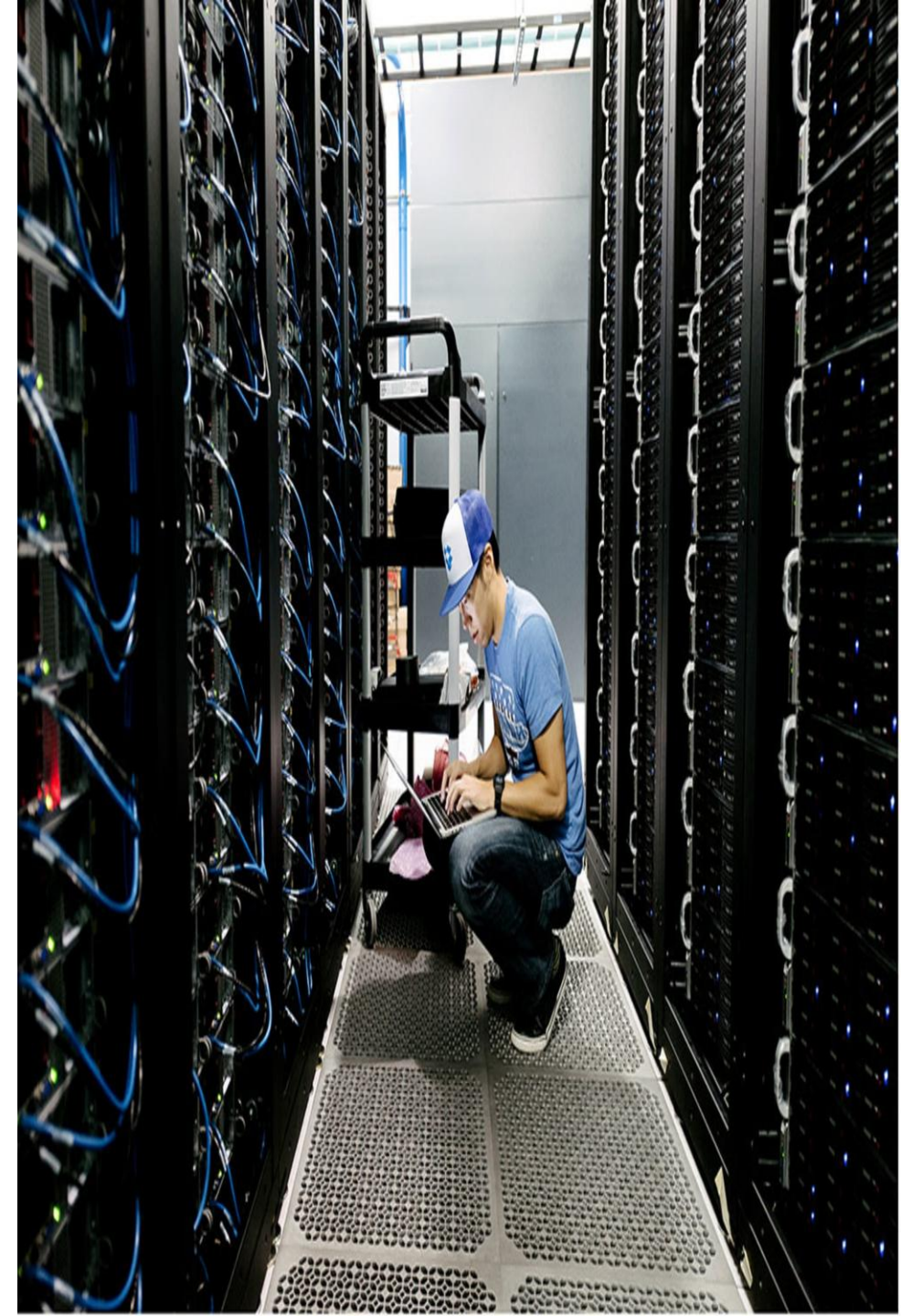
- **Defining Culture**
 - Behavioral Models
 - Organizational maturity models
 - Target Operating Models
- **Improved workflows, communication and feedback loop.**
 - OKR- Metrics and Measurement
 - Immersive, Experiential Learning
 - Innovation - RFC
 - Sharing, Shadowing and Evolving Changes
 - Content collaboration Tools : TFS, JIRA, Wiki, Confluence
 - Single source of truth
- **Collaborative Platforms**
 - Collaboration Model “Where Work Happen” for ultimate DevOps culture
 - On Same Page “Common Hub: Your Team and your hub “
 - Tool:Slack
- **Applying and scaling DevOps in an enterprise environment: 5 Stages of Devops Evolution**
 0. Build The Foundations
 1. Normalize the Technology Stack
 2. Standardize and reduce variability
 3. Expand Devops Practice
 4. Automate Infrastructure Delivery
 5. Provide Self service Capability



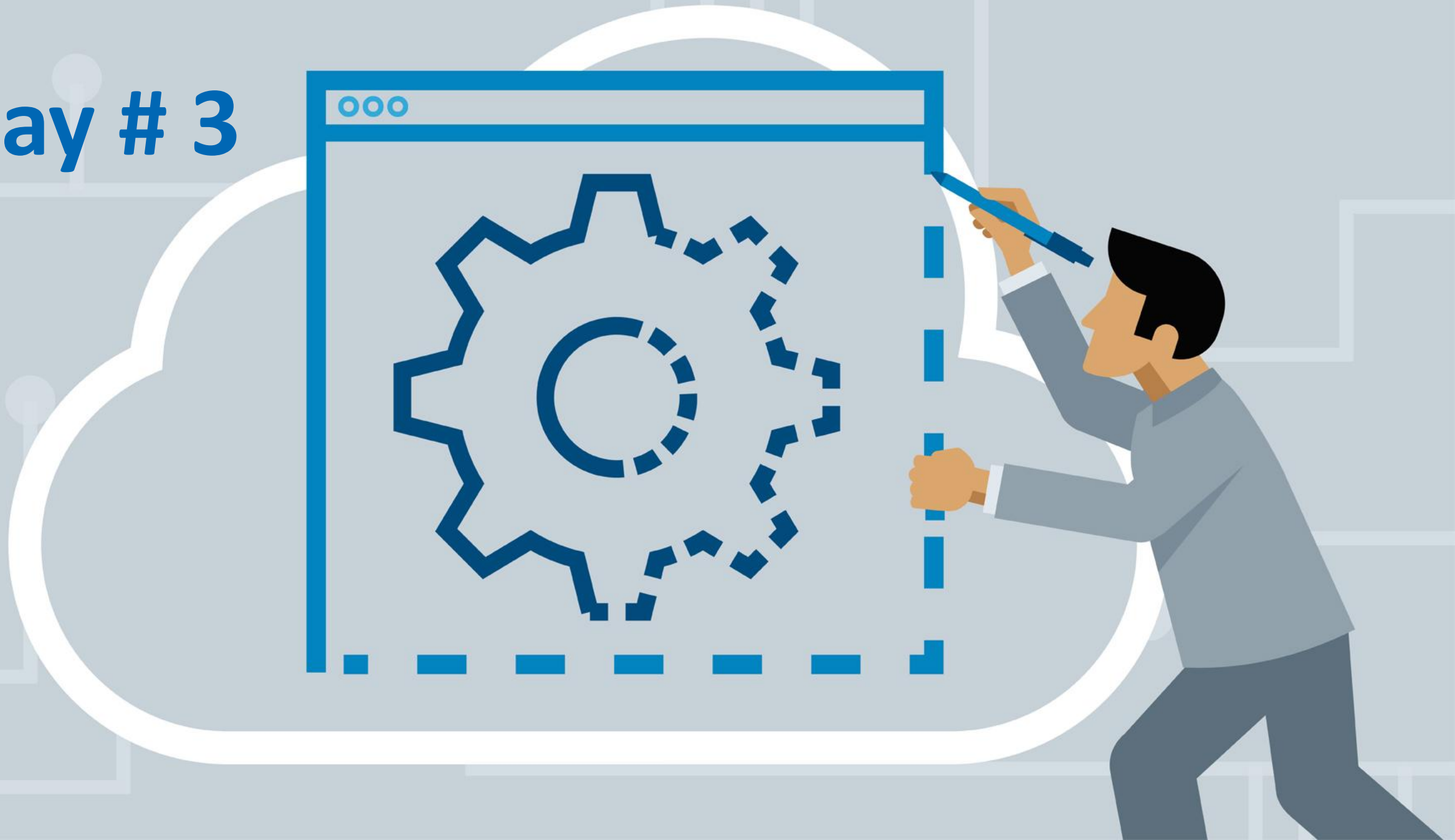
Tech Operations

Infrastructure /Application Performance Management, Monitoring & Support- Operations (SUMO):

- Feedback loop
- APM Tools Configuration
 - New Relic , App Dynamics (Demo and Configuration)
- Smart Alerting Tools
 - Alerta (Demo)
- Dashboard and Monitoring
 - Prometheus (Demo and Hands-on Lab)
- Escalation Tools (VictorOps, Pagerduty)
- 24 7 Support SLA
 - Infrastructure level support
 - Server level support
 - Application level support
 - Add-ons / plugins support
 - Security support
- Incident Management Process
- Case Study: Real Case study for existing running product to be discussed to covered as an exercise.



Day # 3



Self Service- Automation

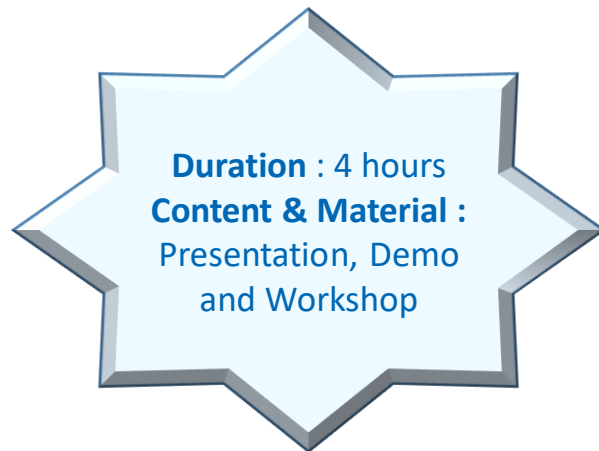


- **Optimize the whole**
- **Perceptions of automation within the organization**
- **Complete Self service (Resources and Ops)**
- Shifting IT capabilities as Service
- Service Management Platform
- Security Policy Automation
- Incident responses automation
- Devops Toolchainning services
- Toolchain orchestration
- Unified process, people and tool
- **Benefits:**
- Efficiency
- Cost
- Trust vs Control

Duration : 2 hours
Content & Material :
Presentation and
Workshop

Cloud Adaption and Implementation

- Introduction
- Cloud construction, Cloud Migration
- Scalability, Reliability and High Availability
- AWS – Services Intro and Usage : Demo
- Azure- Service Intro and Usage : Demo
- Optimization Techniques
- Cloud Governance
- Benefits



DevSecOps



- Turning DevOps into DevSecOps
- Seamless Integration with Security
- Enable security to move at the speed of digital business
- Interactive Application Security Test (utilize AST Tools) in Development
- Runtime Application Self-Protection Testing in Production
- Lightweight Application Security Testing
- Scan All Developer Constructs for Vulnerabilities and Proper Configuration
- VMs, AMIs, containers are all styles of "containers"
- Hardening Workshop
- Scan for known vulnerable software and proper configuration:
 - Operating Systems
 - Application platform
 - OSS and commercial off-the-shelf software

Duration : 2 hours
Content & Material
: Presentation and
Demo

An Introduction to Agile & Scrum



Training Objective

- To train people on Agile methodology and Scrum process, make them enable for Agile, tell them the real philosophy of agile and scrum, so everyone will be on same page so we can get maximum benefit of this framework.
- To provide in depth understanding and thought process of Scrum artifacts, Scrum Roles, Scrum ceremonies , Sprints , Story points so one can relate that to daily activity.

Overview of Training

Hour 1

- Why Agile?
- Background
- Agile Manifesto
- Agile Principles

Hour 2

- Introduction to Scrum
- Roles
- Artifacts
- Ceremonies

Hour 3

- Practical look at Scrum
- Story Points

Agile: An introduction



Overview of Hour #1

- Background
 - Why Agile?
 - Introduction
 - History of Agile
- Agile Manifesto
- Agile Principles

Background

Waterfall Method

- Waterfall method approach is a unidirectional (Moves in one direction) approach to SDLC. You go to next phase when previous phase is complete. Once you go to second step you can not go back to first step.

- SDLC

Requirement
Gathering



Analysis



Design



Implementation

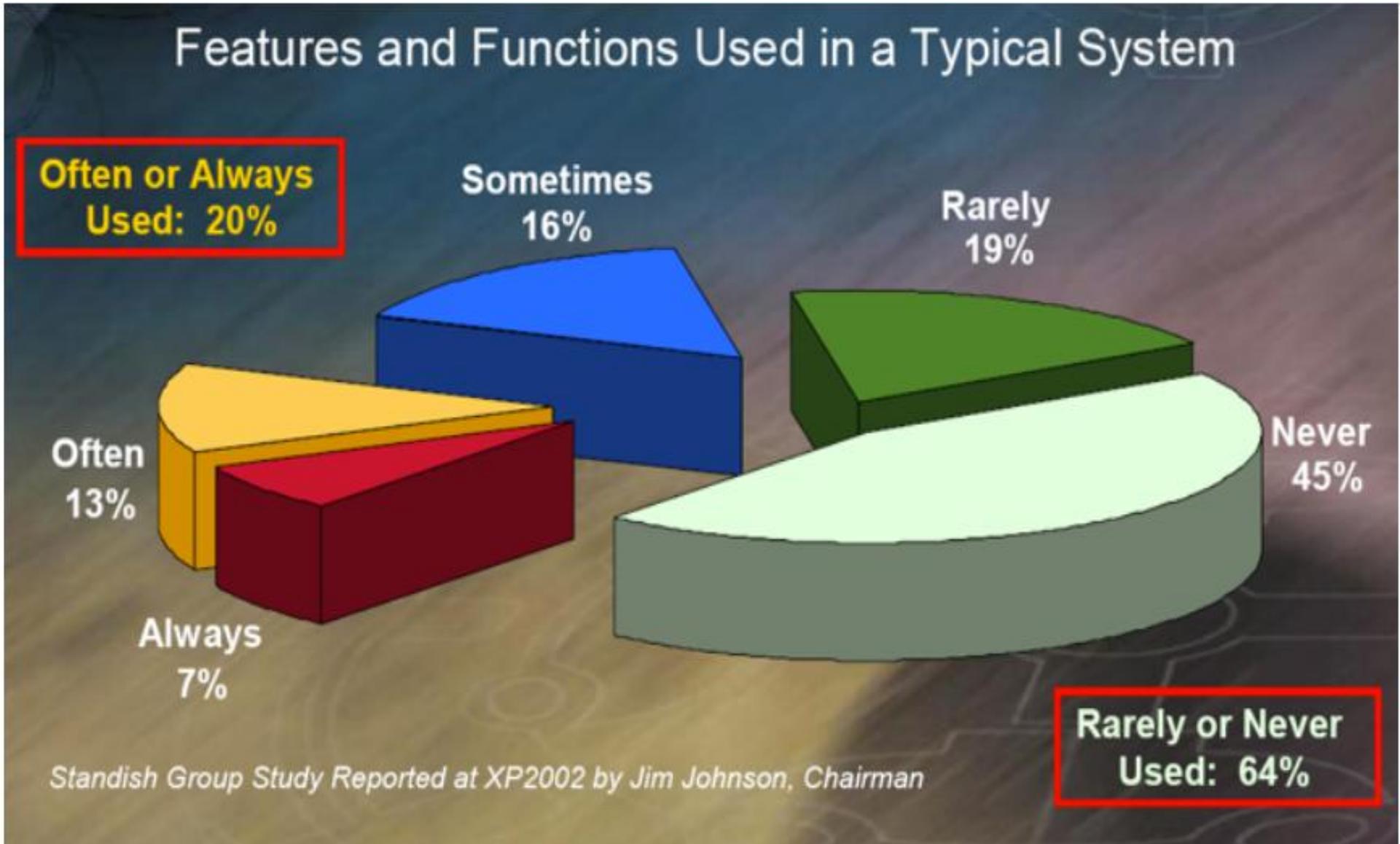


Testing



Deployment

Study Report

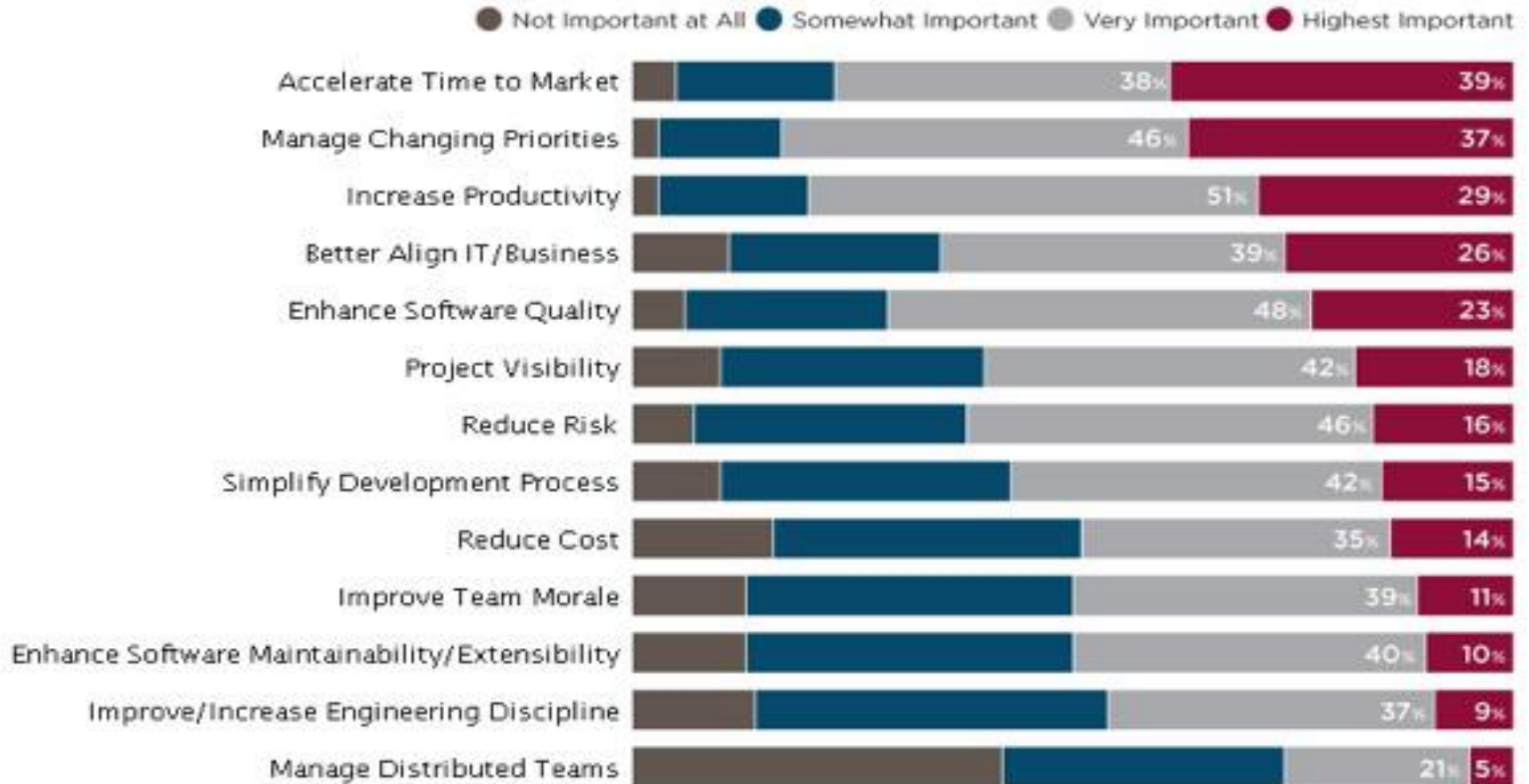


Need for Change

- Is traditional process working? Why?
- What are some typical issues faced in Traditional projects?
- What do we typically do when we find out that our project is late?

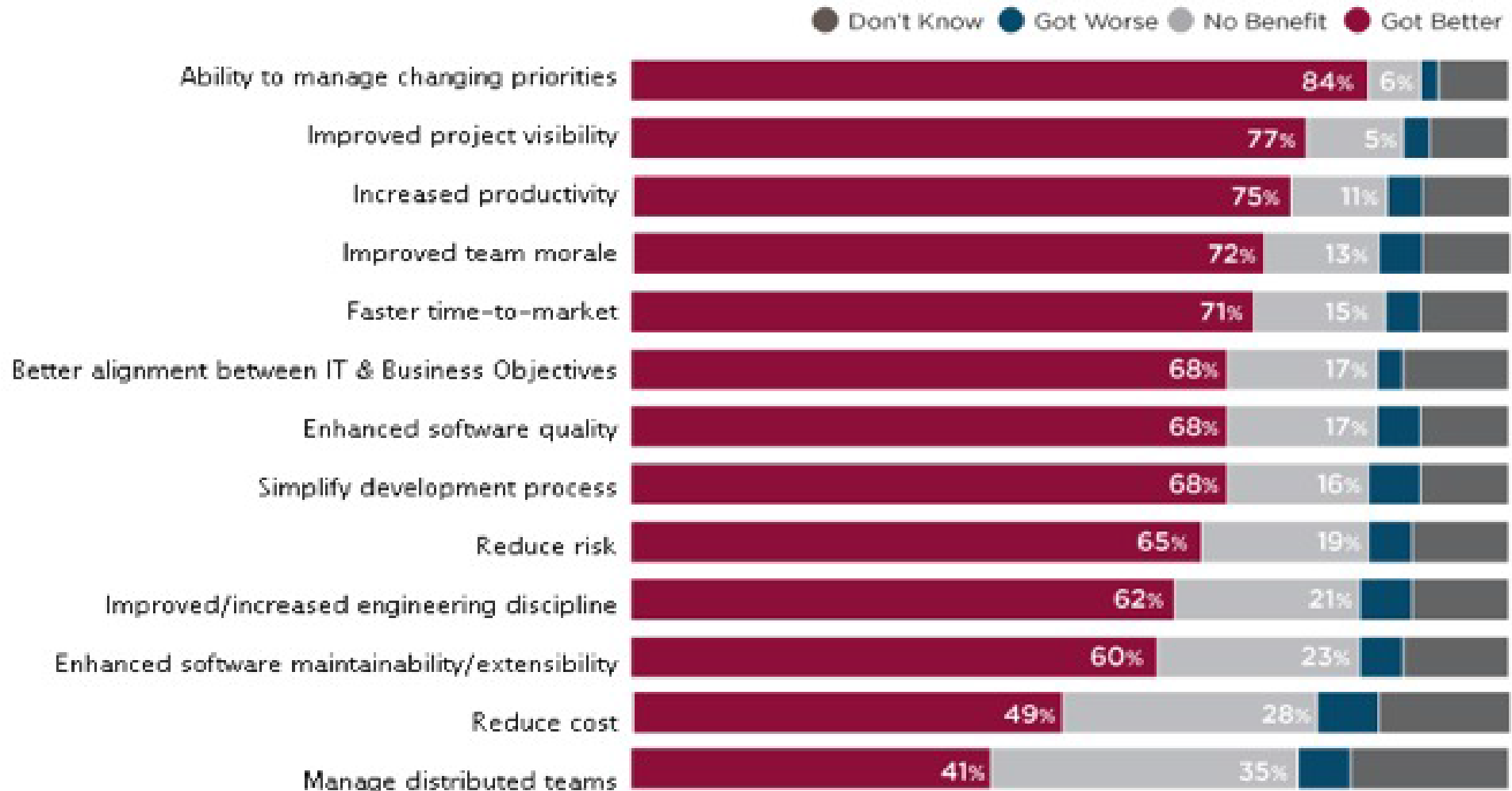


Why Agile?



The most common reason for adopting Agile?
Accelerate Time to Market

Why Agile? What are the getting



the greatest benefits from the **ability to manage changing priorities**.

History

1960 1970 1980

1976

Tom Gilb



Adaptive iterations,
fast time to market

1985

Barry Boehm



Spiral Model
Priority based on risk

1950s -1960s
X-15 hypersonic jet



Iterative Incremental
Delivery According to
Craig Larman and Victor
Basili*, IID grew from
the work of Walter
Shewart in the 1930s.

1980

Gerald Weinberg



Small Increments,
Customer-driven
feedback

1986

Fred Brooks "No
Silver Bullet" and
"Mythical Man Month"



1985

Hiroataka Takeuchi &
Ikujiro Nonaka



**The New New Product
Development Game**

- Self-organizing team
- Cross-functional team
- Empowered team



1990



1990 - Sutherland & Schwaber
Scrum PM Framework
Time-boxed iterations (30 days)
Small and co-located, Inspect & adapt



1995 – Booch,
Rumbaugh & Jacobson
Rational Unified Process
Architecture Focus



1996 - Beck,
Cunningham & Jeffries
Extreme Programming
Engineering Practices

1994

DSQM
CONSORTIUM

**Dynamic System
Development Method**
Formalization of RAD



1997 - Jeff de Luca
Feature Driven Development
Deliver tangible, working software
repeatedly in a timely manner



1998 - Alistair Cockburn
Crystal Family
People & Communications, Design
Principles, Domains, Bare Sufficiency



2000 – Robert Charette
Lean Development
Strategic Focus, Lean Production,
Risk Entrepreneurship,
Stretch Goals

2000



Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals & interactions	over	Processes & tools
Working software	over	Comprehensive Documentation
Customer collaboration	over	Contract negotiation
Responding to change	over	Following a plan

That is, while there is value in the items on the right,
we value the items on the left more.



Agile is not...

- New
- A silver bullet
- Without planning or documentation
- An excuse for poor quality
- Undisciplined
- Unproven

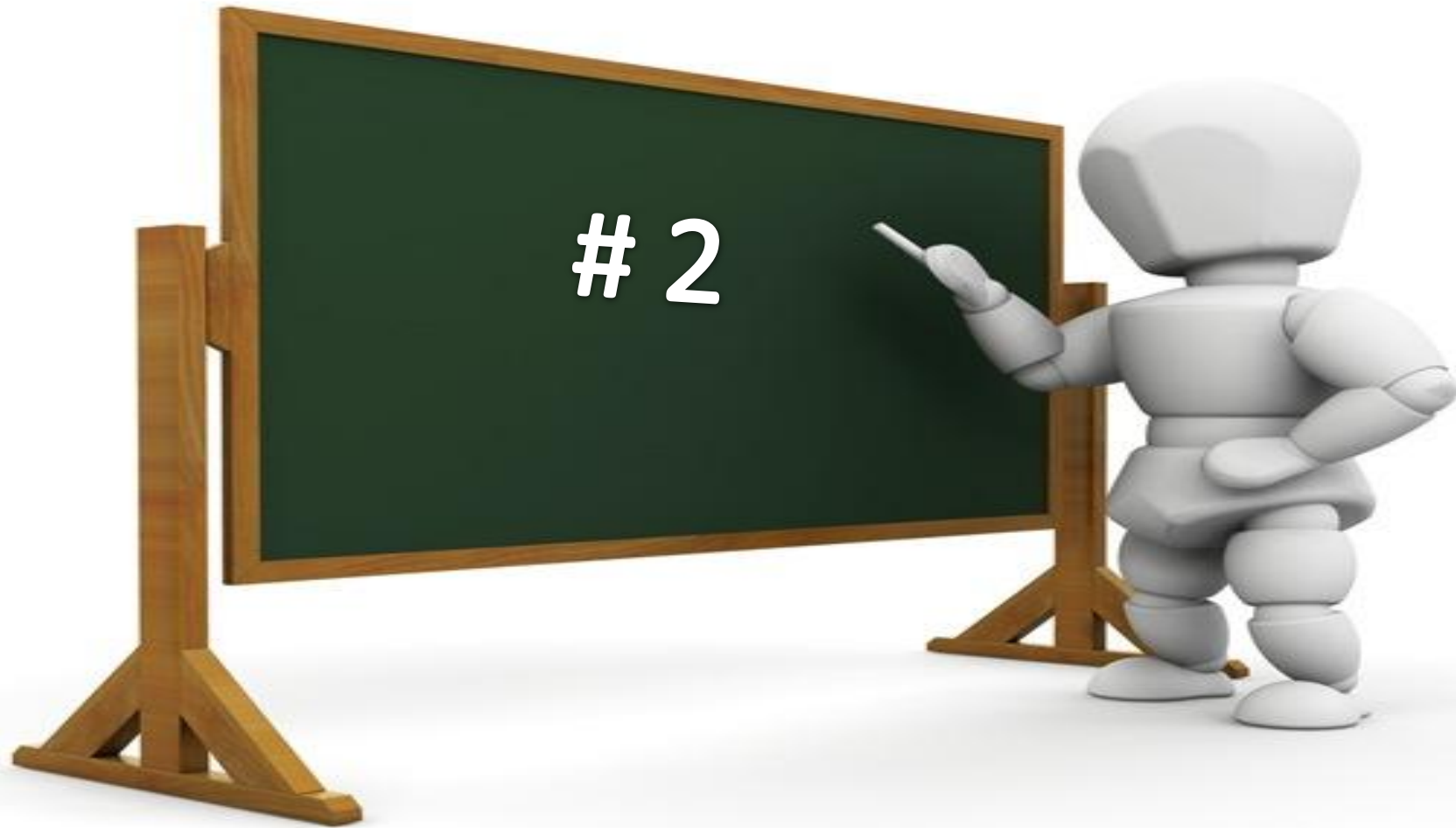
12 Principles of the Agile Manifesto

1. Produce Value Early
2. Welcome Change
3. Interactive Delivery
4. Daily Business Collaboration
5. Trust Motivated Team
6. Face-to-Face Conversations
7. Working Software
8. Sustainable Pace
9. Technical Excellence
10. Simplicity
11. Self-Organize
12. Reflect

Exercise



Scrum: Introduction



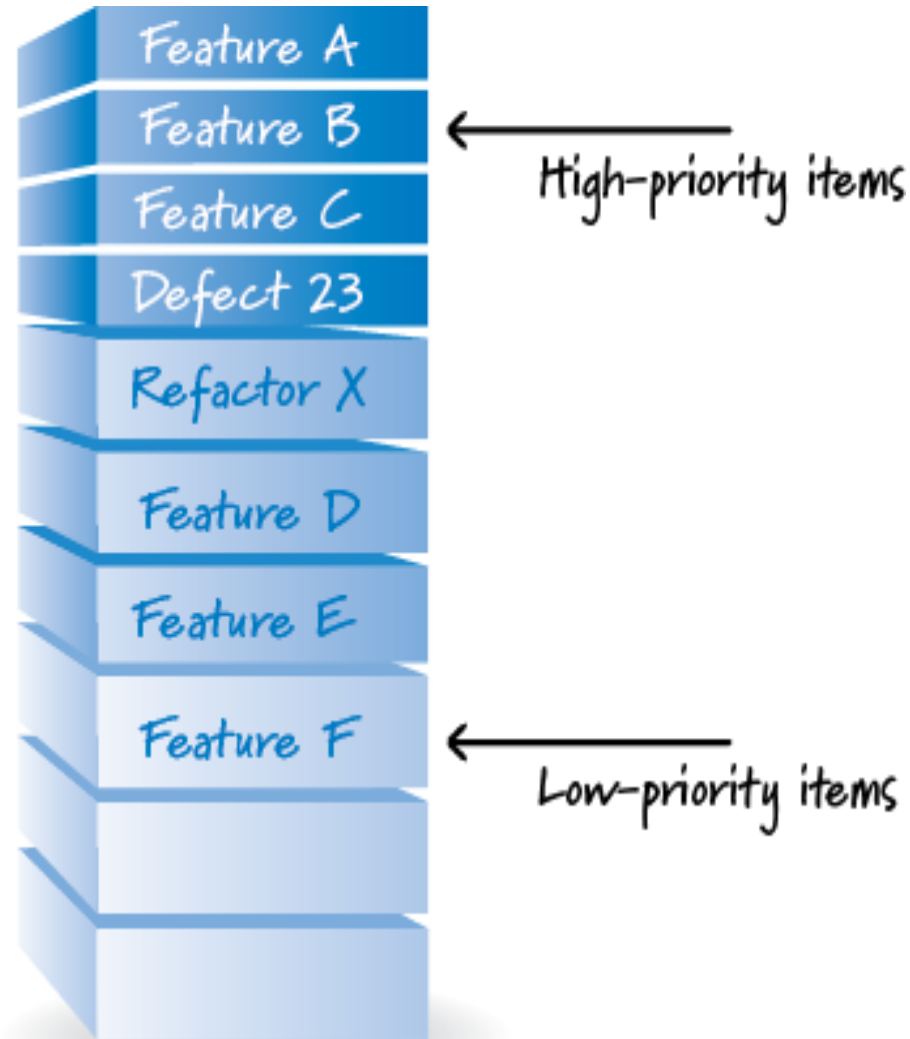
Overview of Hour # 2

- Scrum
- Scrum Flow
- Scrum Roles
- Scrum Artifacts
- Scrum Ceremonies
- Scrum Values
- Build your own Scrum

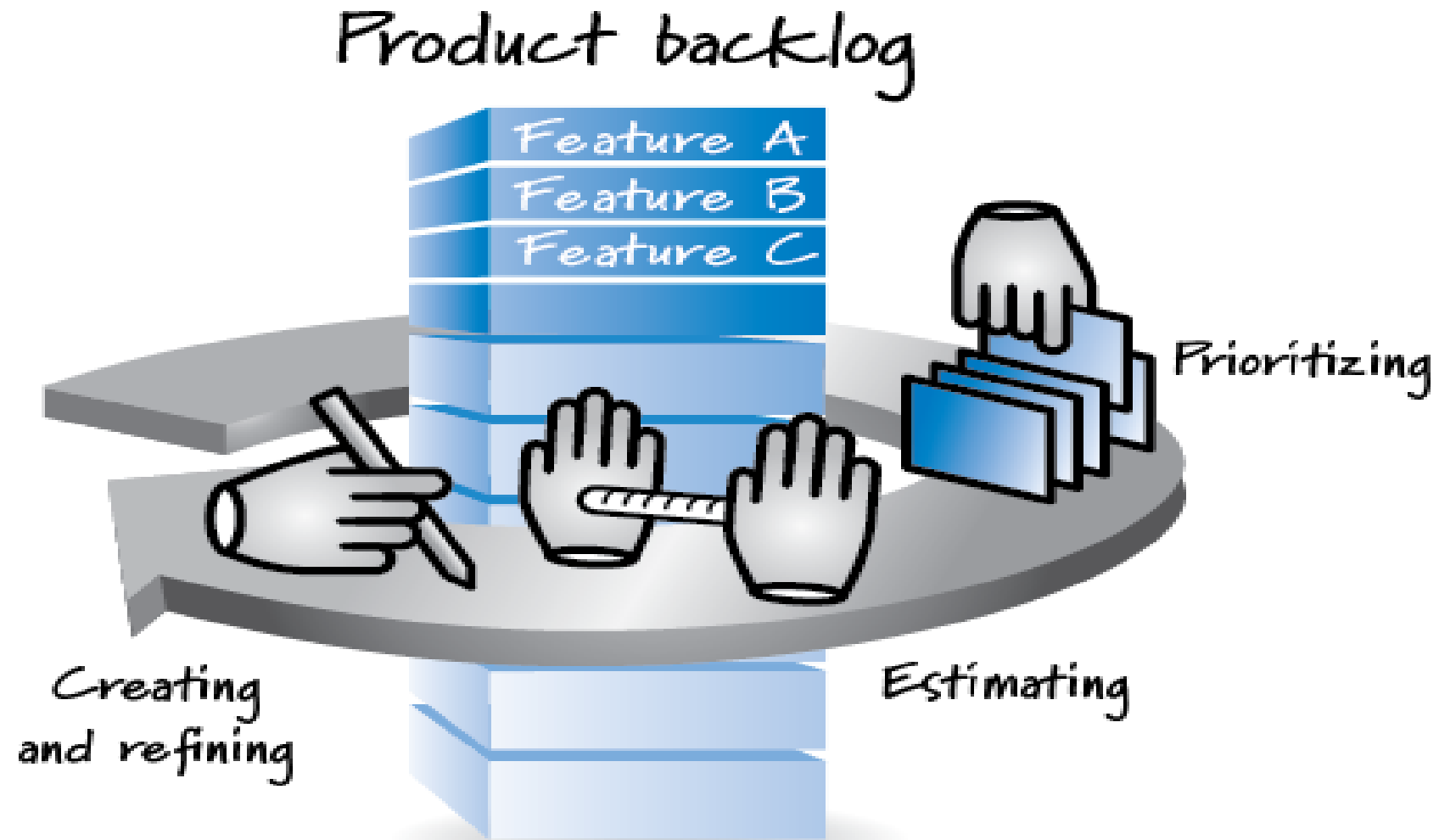
Scrum Flow :Idea



Scrum Flow: Product Backlog



Scrum Flow: Grooming



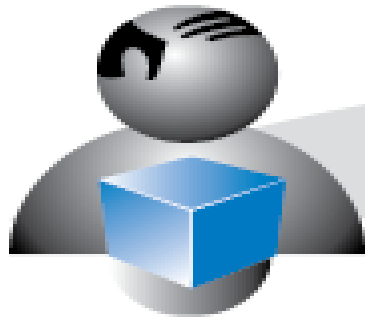
Definition of Done



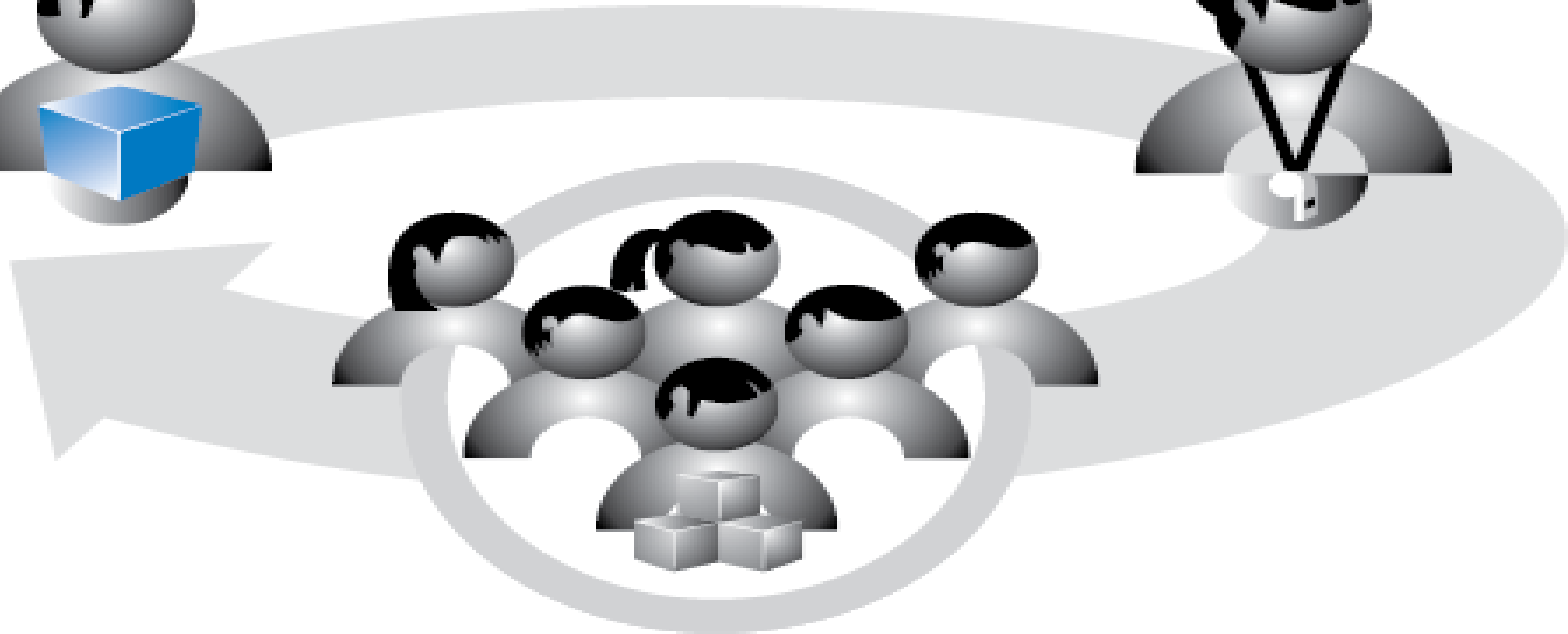
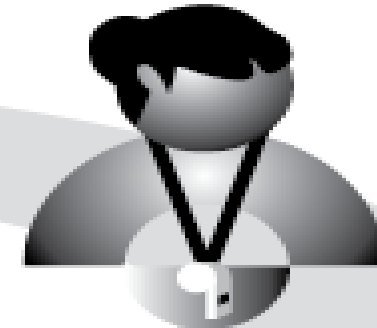
Scrum Flow: Scrum Roles

Scrum team

Product owner

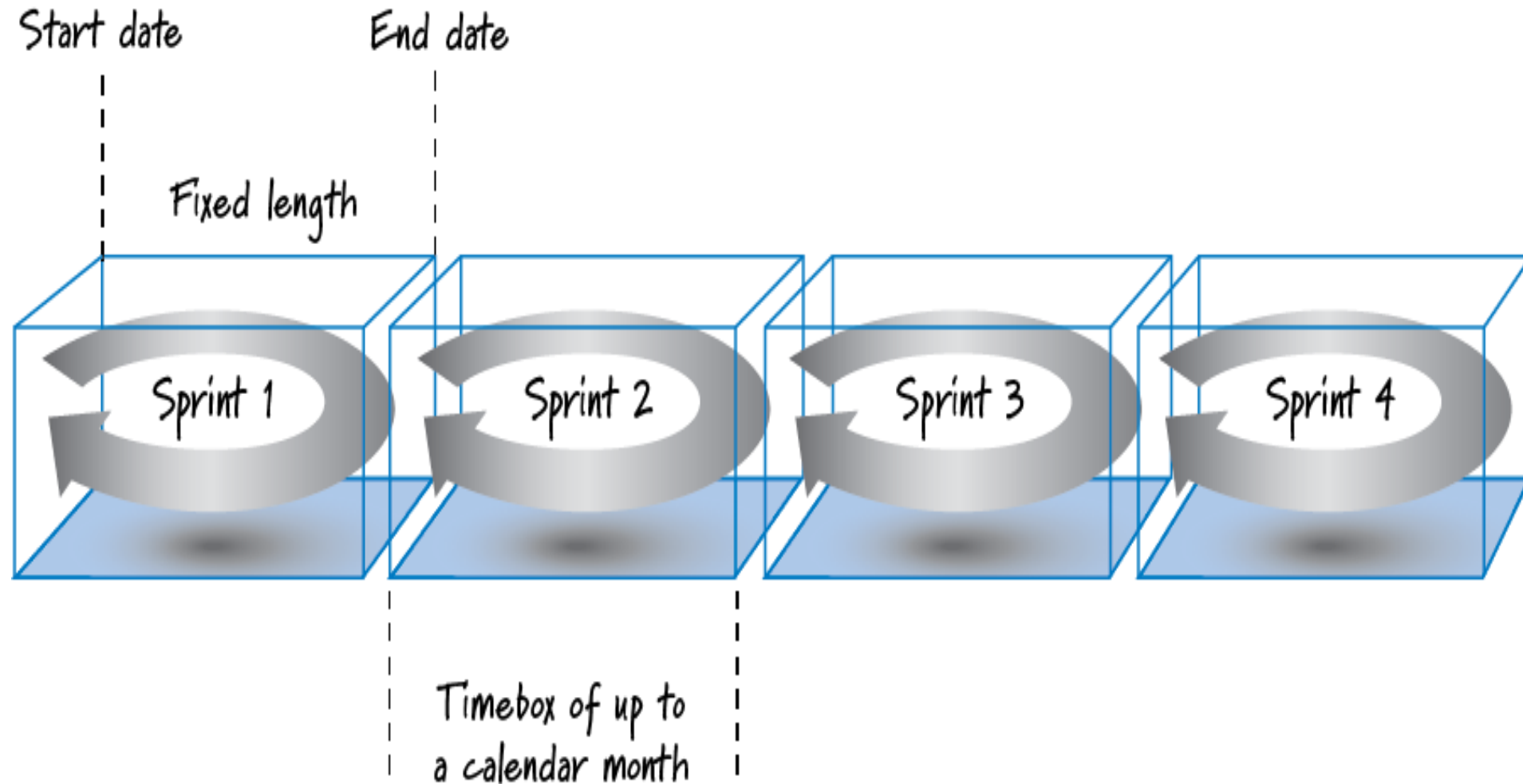


ScrumMaster

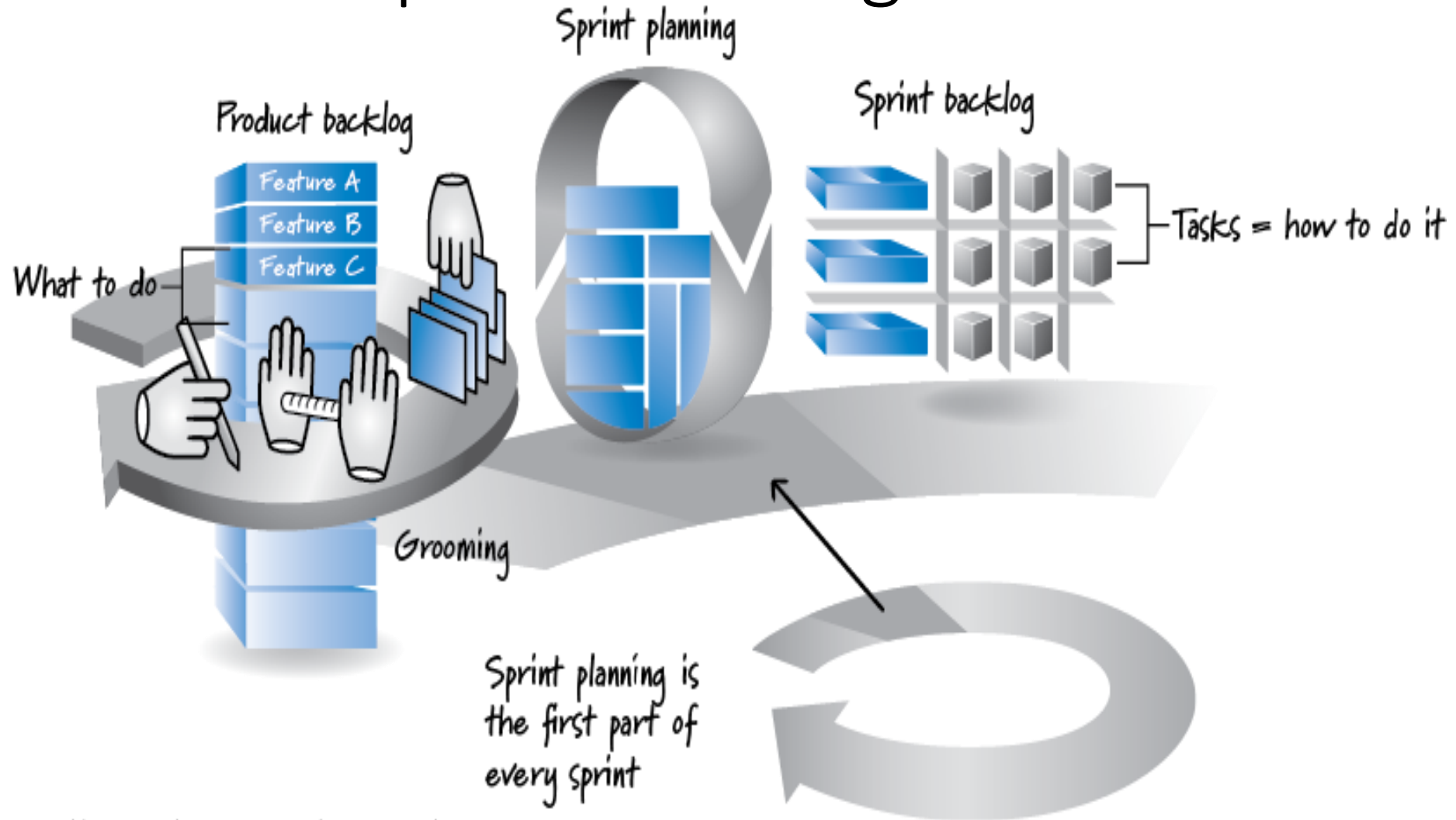


Development team

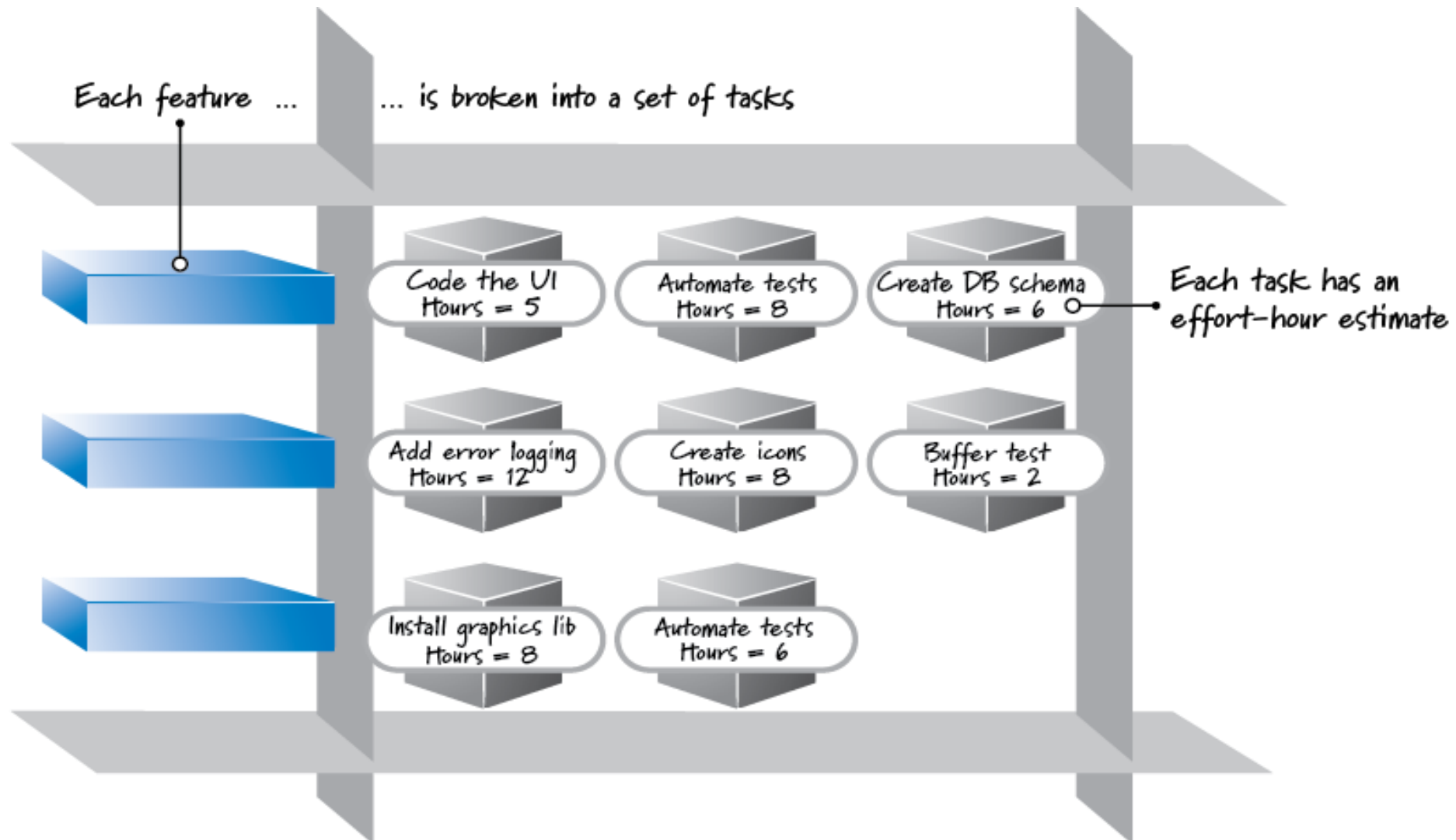
Scrum Flow: Sprint



Scrum Flow: Sprint Planning

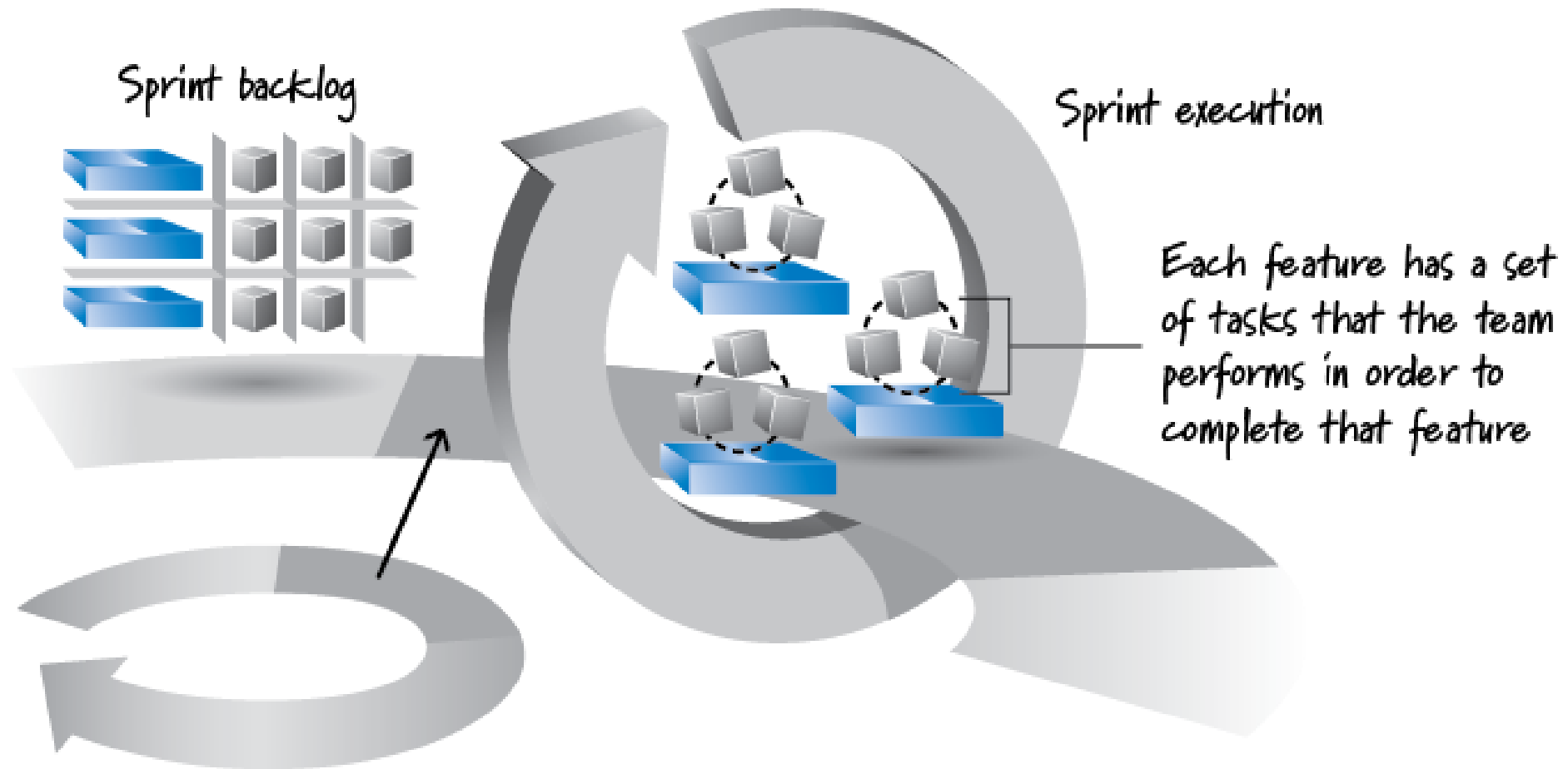


Scrum Flow: Sprint BL

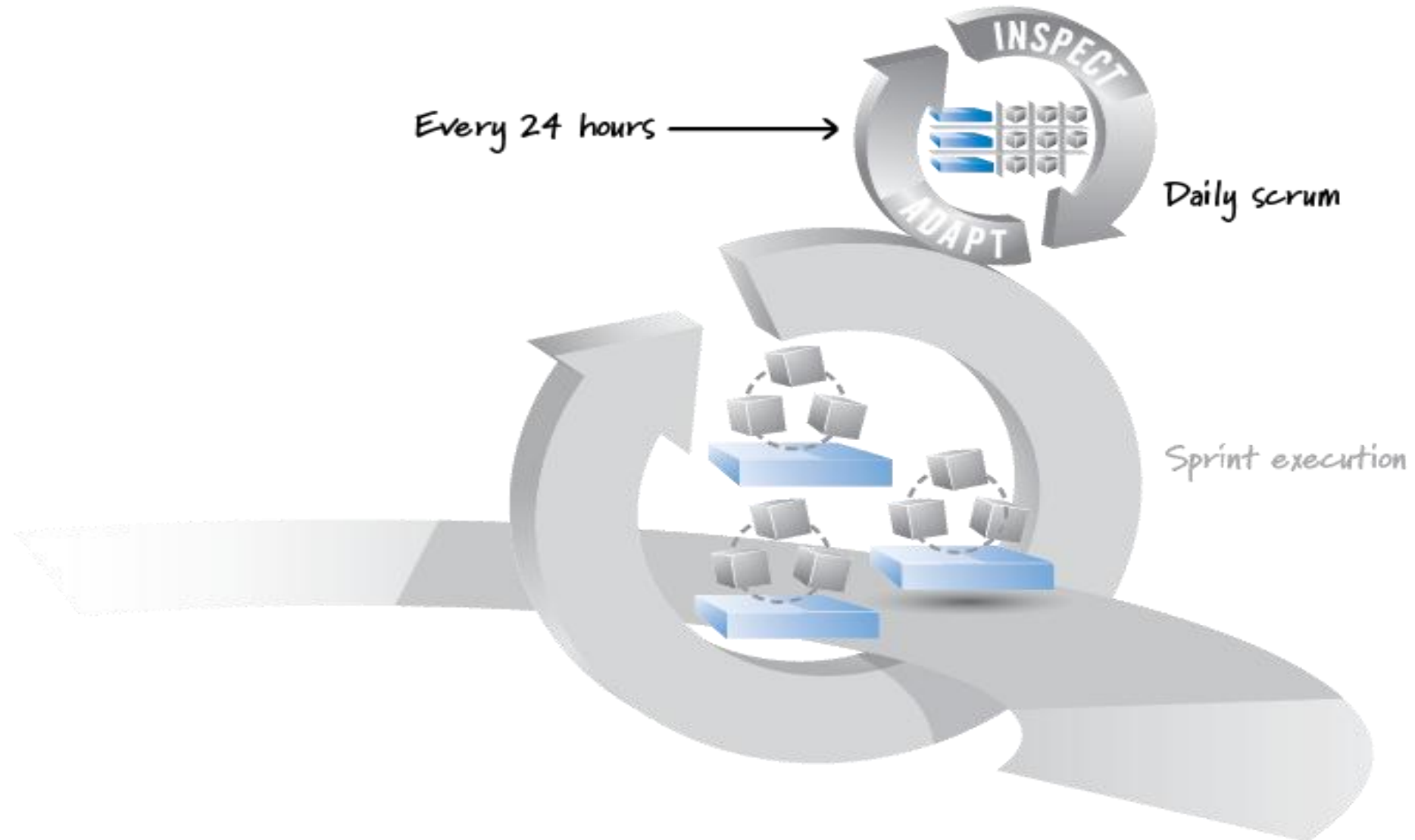


Scrum Flow: Sprint Execution

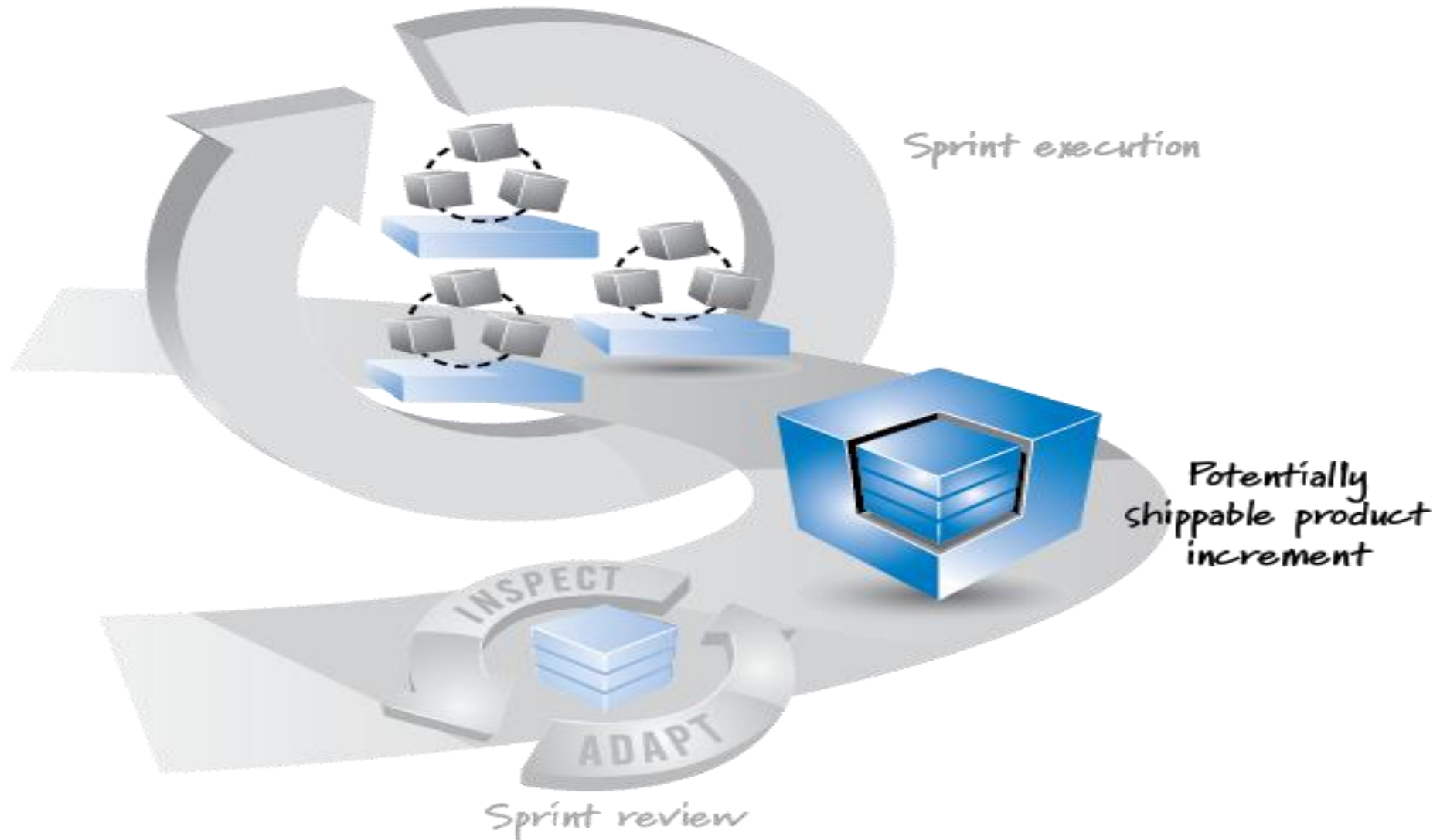
Sprint execution takes up the majority of time spent in each sprint



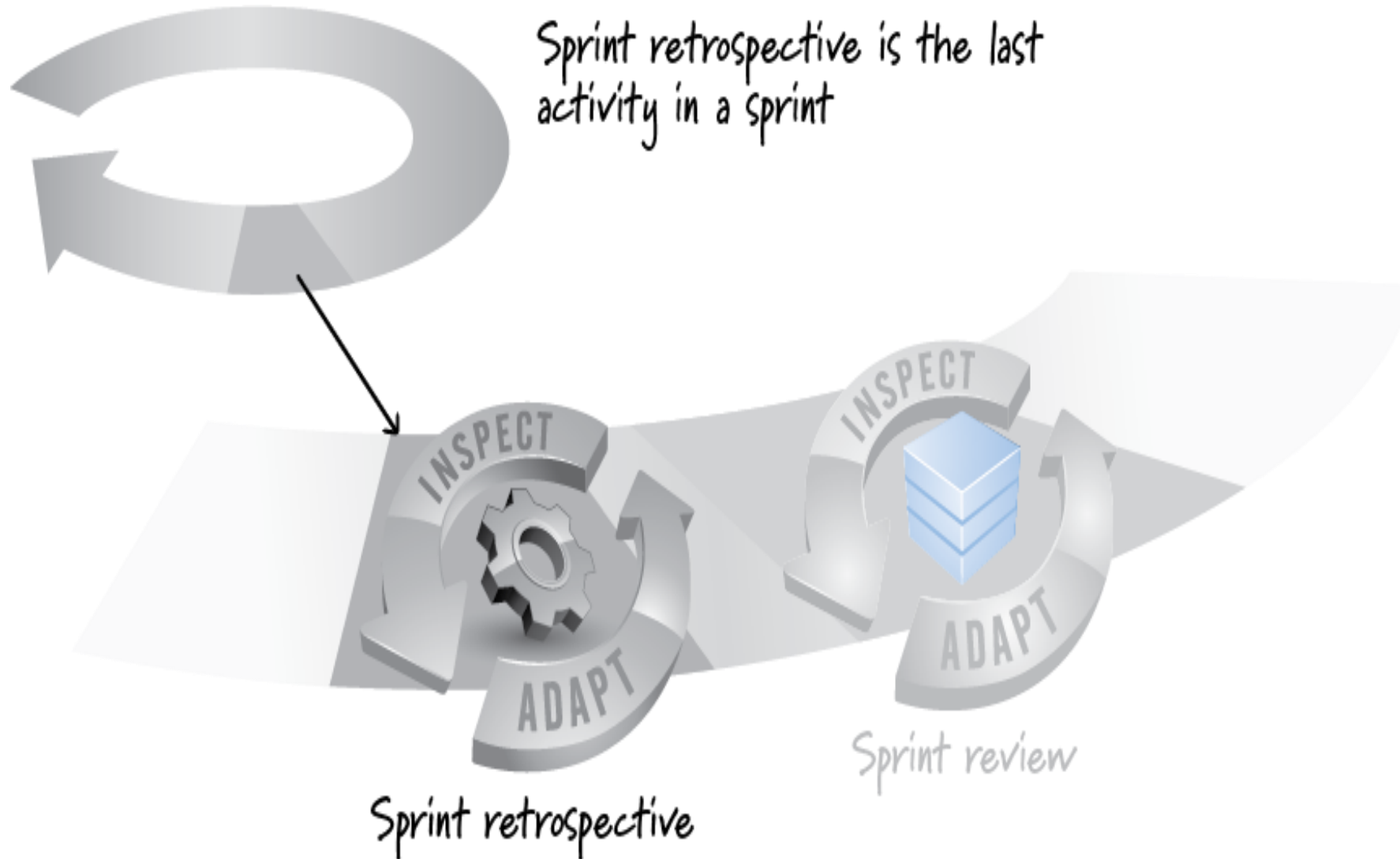
Scrum Flow: Daily Scrum



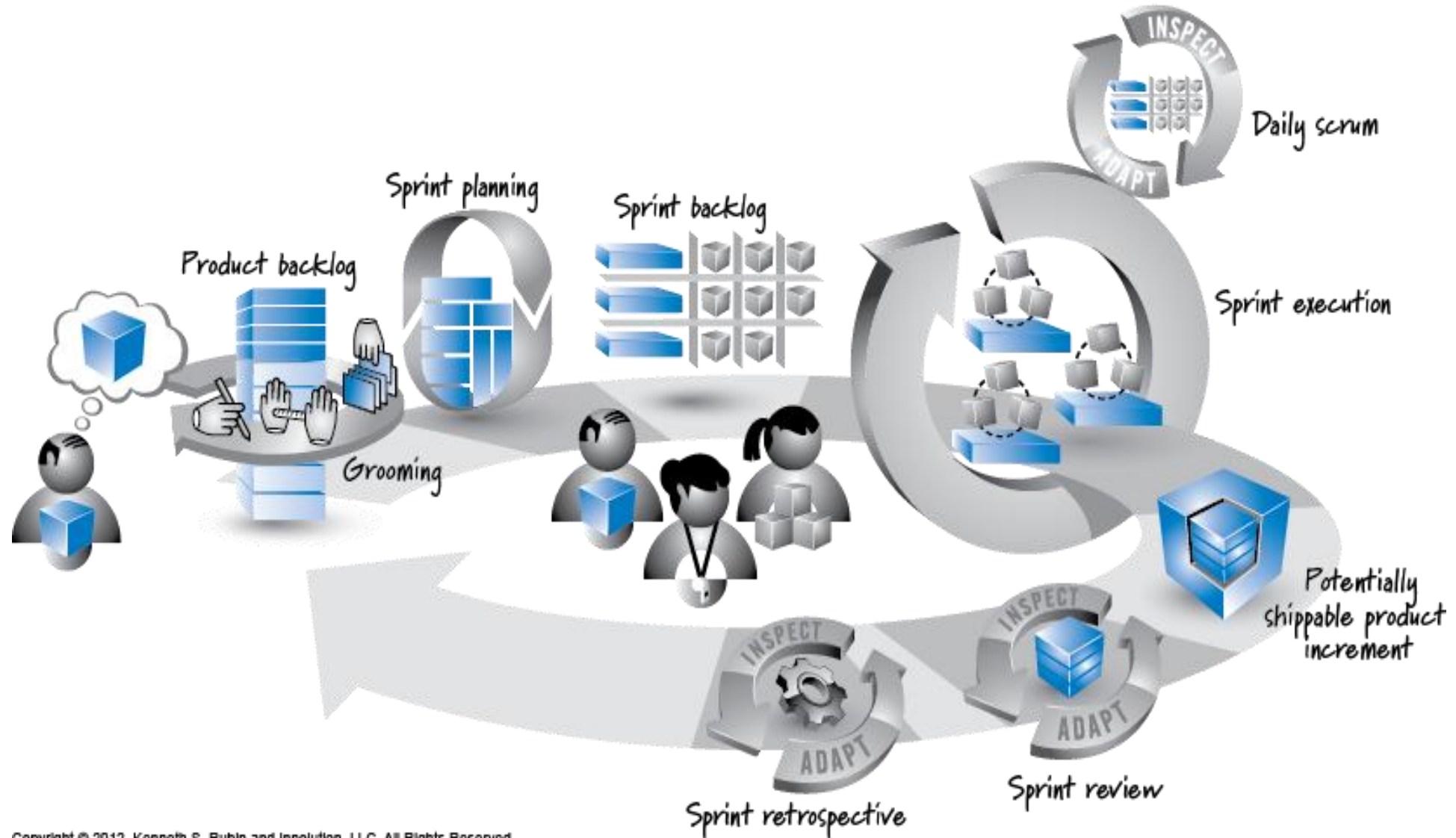
Scrum Flow: Sprint Review



Scrum Flow: Sprint Retrospective



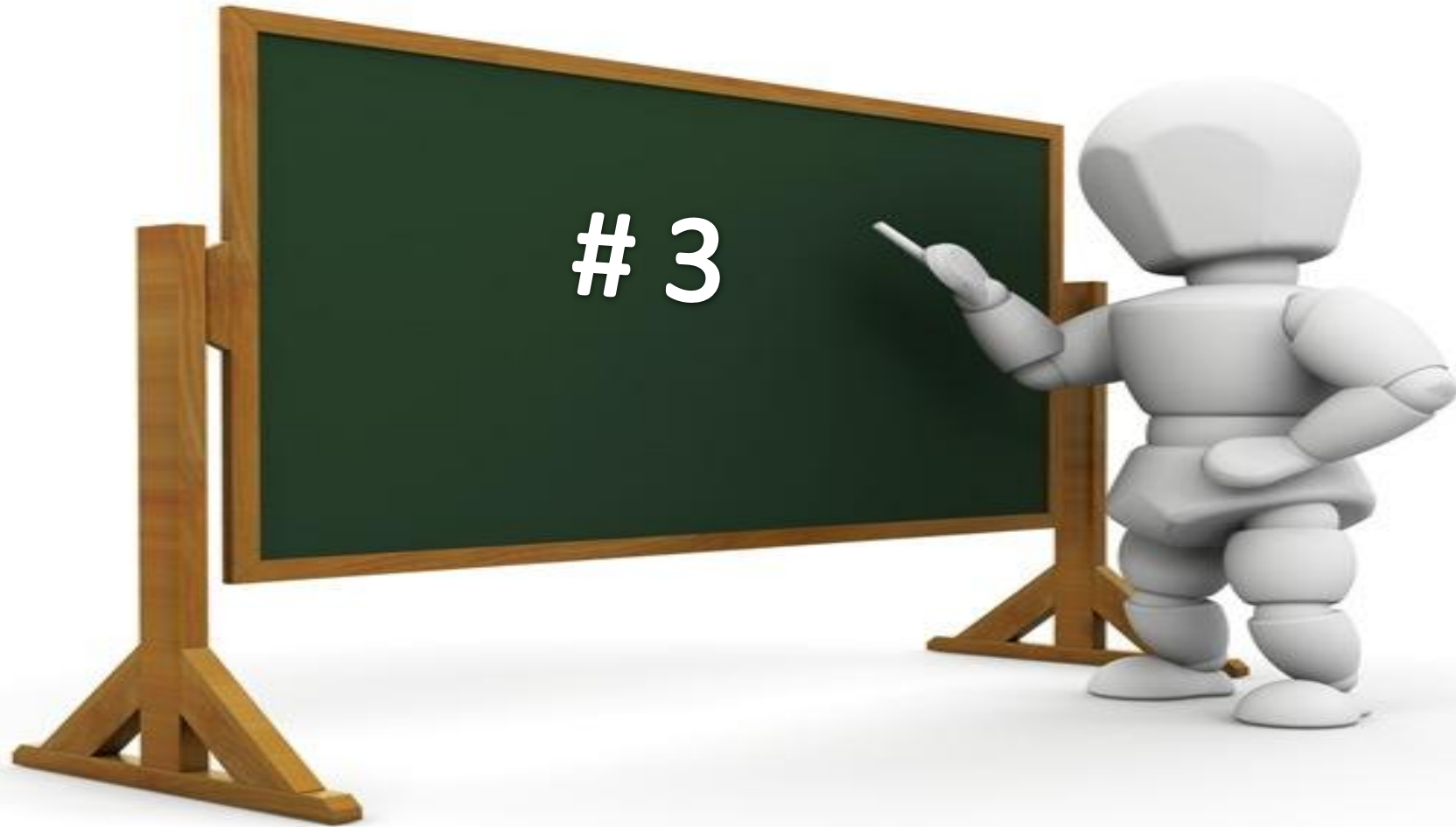
Scrum Flow



Build your own Scrum



Scrum: Practitioner approach



Overview of Hour # 3

- User Stories
- Story Points
 - Estimating Techniques
- Velocity
- Planning Poker
- Day 2 (Spill over)
- Conclusion

User Stories

Written in a standard format

- As a <who>, I want <what>, so that <why>
- As a user, I want to search for my customers by their first and last names so that I can access them quickly.

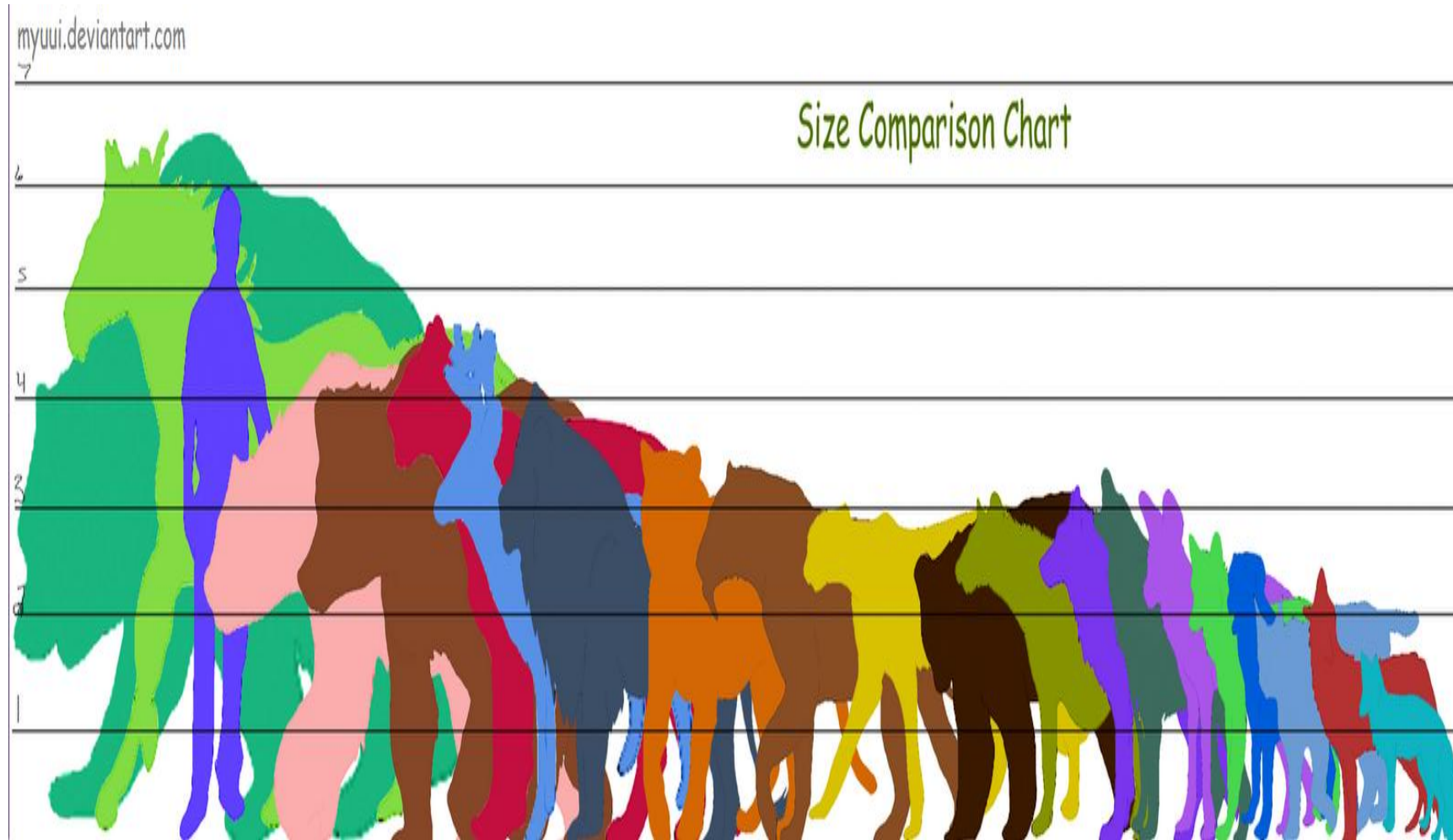
Story Points

- Based on complexity, Effort and Uncertainty
- Relative Value
- Consensus based technique

Estimation Techniques



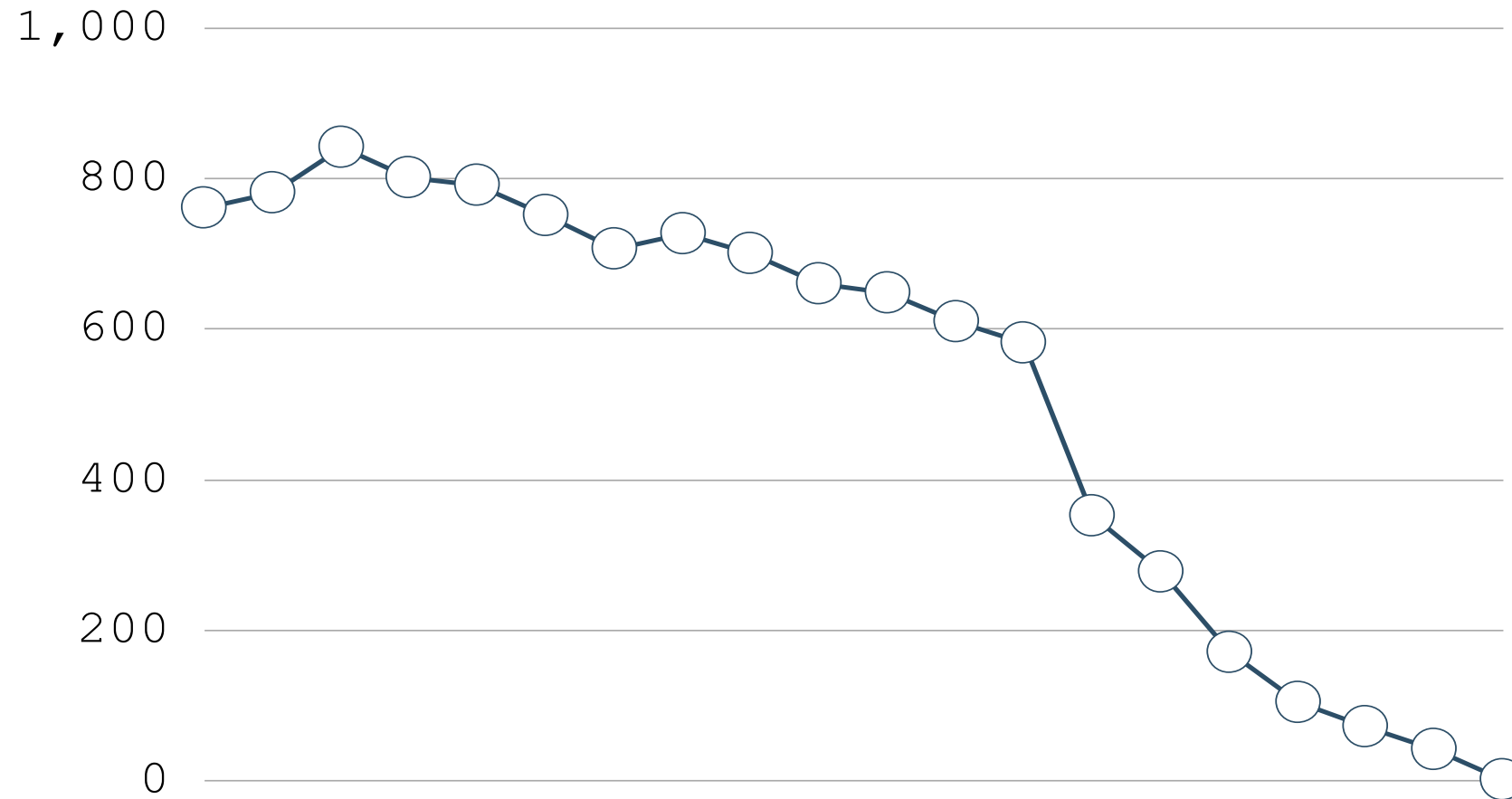
Estimation Techniques



Planning Poker



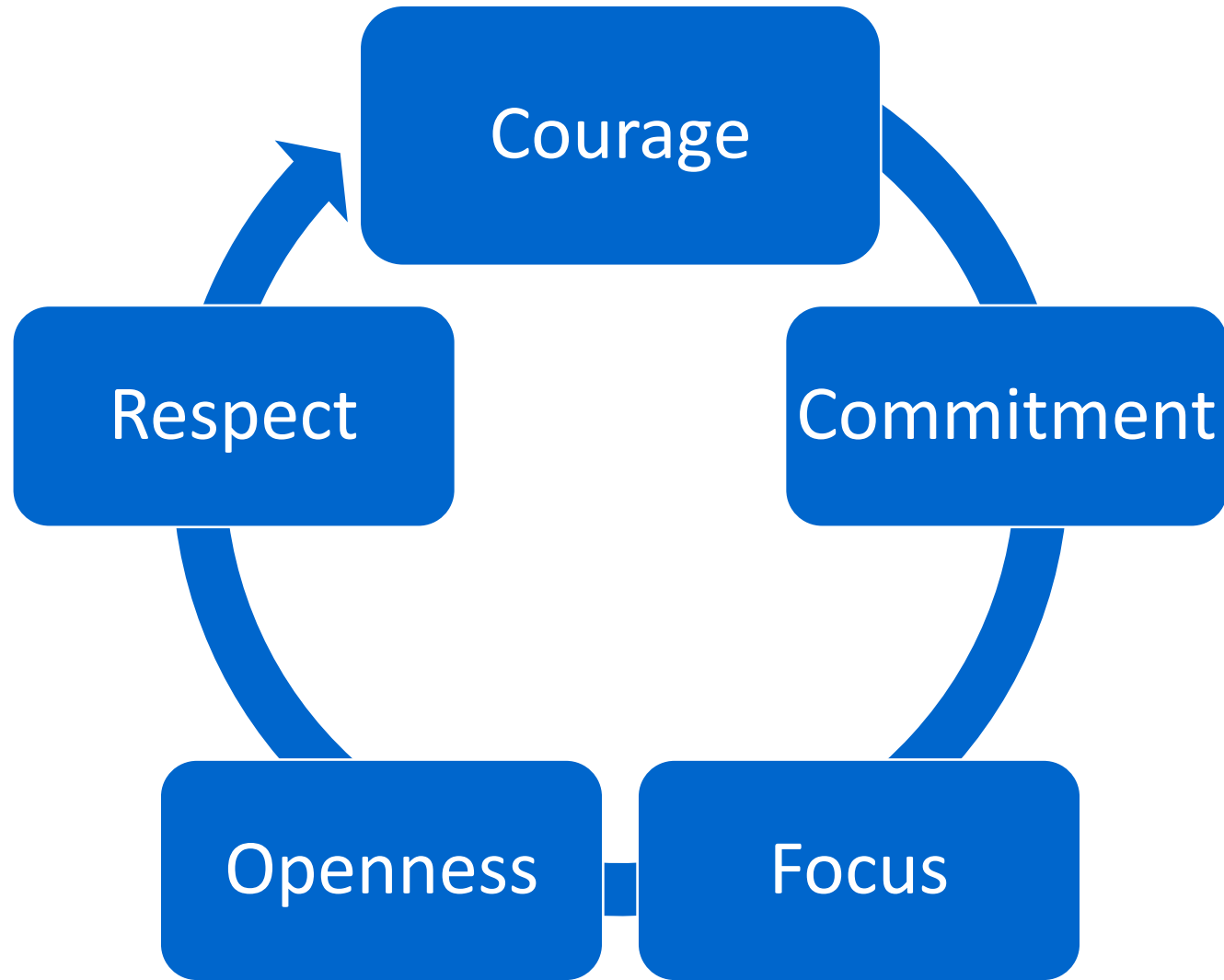
Burn-down chart



Velocity

Velocity = No. of Story Points achieved per sprint

Scrum Values



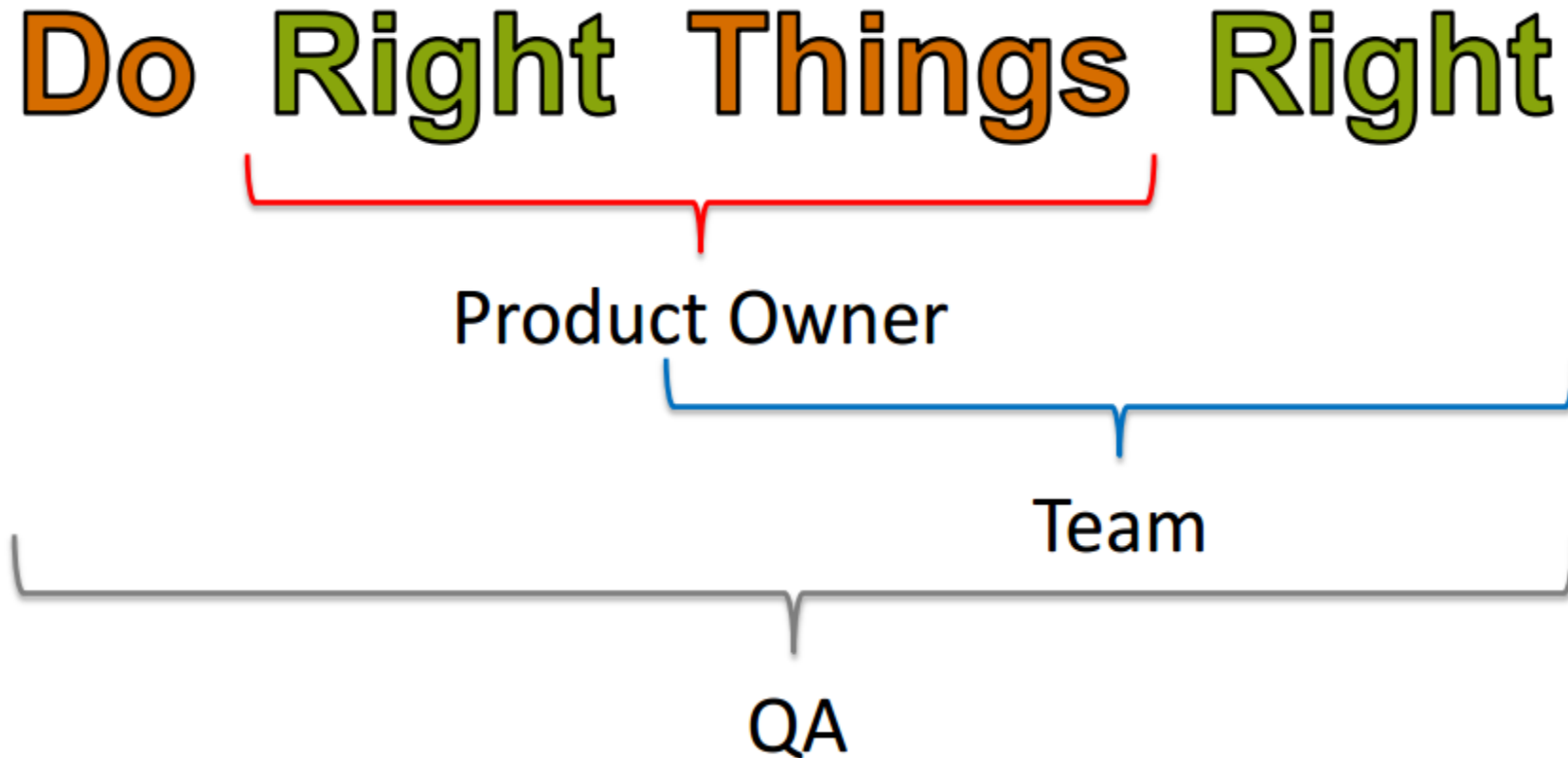
“Try not to become a man of success. Rather become a man of value.”

- ♦ Albert Einstein

Scrum Terminologies

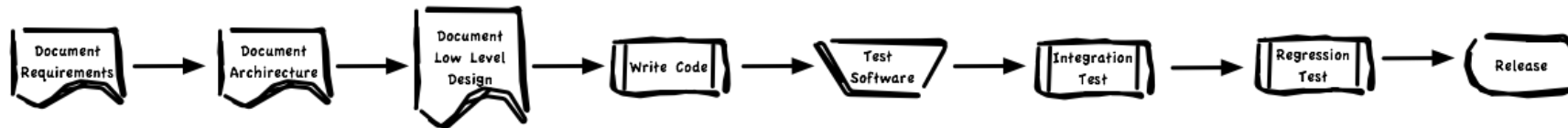
- Product Owner
- Scrum Master
- Scrum Team
- Product Backlog
- Sprint Backlog
- Sprint Planning
- Daily Scrum
- Sprint Review
- Sprint Retrospective
- Product Increment
- Sprint burn down chart
- Release burn down chart
- (User) Story
- Theme
- Epic
- Definition of Done (DoD)
- Velocity
- Impediment
- Point Scale/ Effort/ Story points
- Planning Poker

QA in Agile

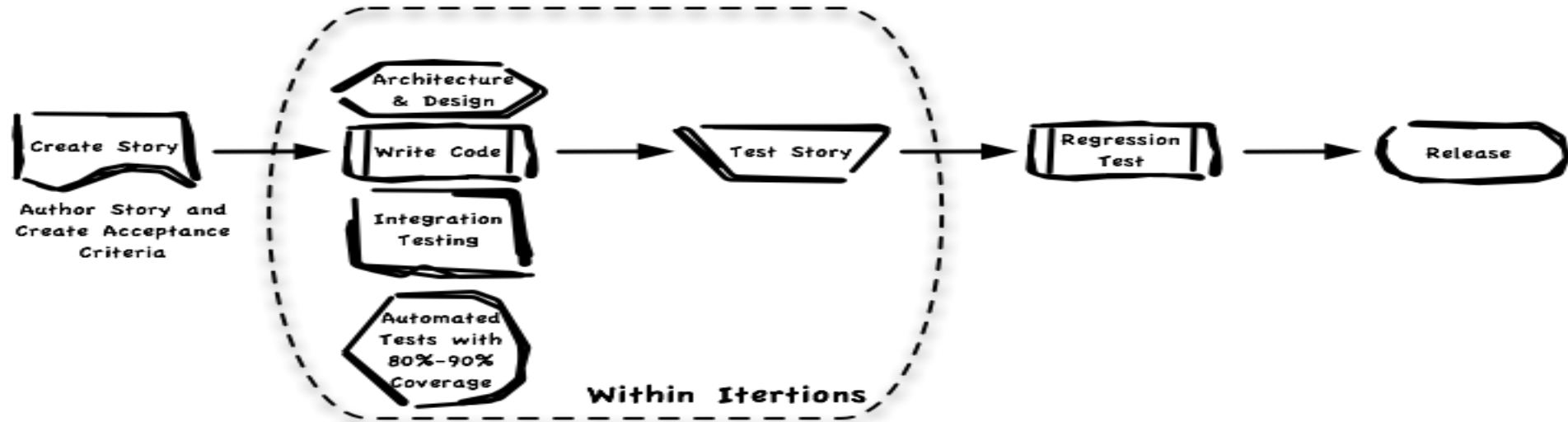


Testing in Agile world

Traditional Testing Approach



Common Agile Testing Approach



Testing Manifesto



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SQA Responsibility in Agile

- Product owner and customer perspective against each story.
- Definition of Ready
- Definition of Done
- Unit Testing framework has been setup and has complete coverage
- Unit testing run is in use and running with CI Server
- QA env setup
- Validation of Story
- System env setup
- Regression automation
- Regression testing happens
- Regression testing coverage
- Continuously update regression testing run as per product evolutions
- Non Functional testing run

Case Study – TPS Pakistan

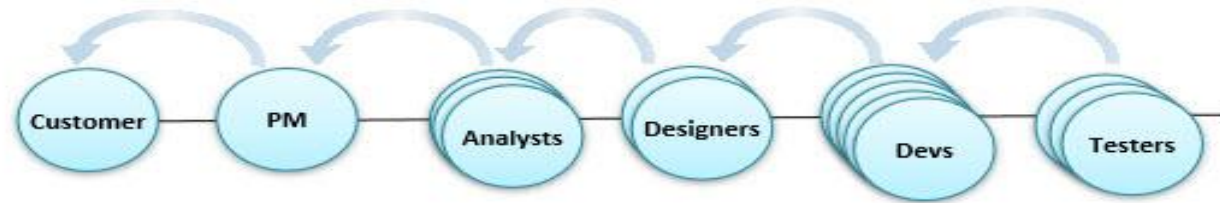
- Same Product – Multiple Code Repos
- New Client – Hire Same number of Team- Same Effort repeat
- Production Deployment 3 to 6 Months Project
- Client itself is Waterfall – How come we be Agile.
- Development Cycle –
 - All Effort are Manual
 - Multiple Testing Cycle
 - Huge Centralize QA team
 - Regression Hell
 - QA environment not Available – Long Cycle
 - Silo Management

Case Study- TPS Pakistan

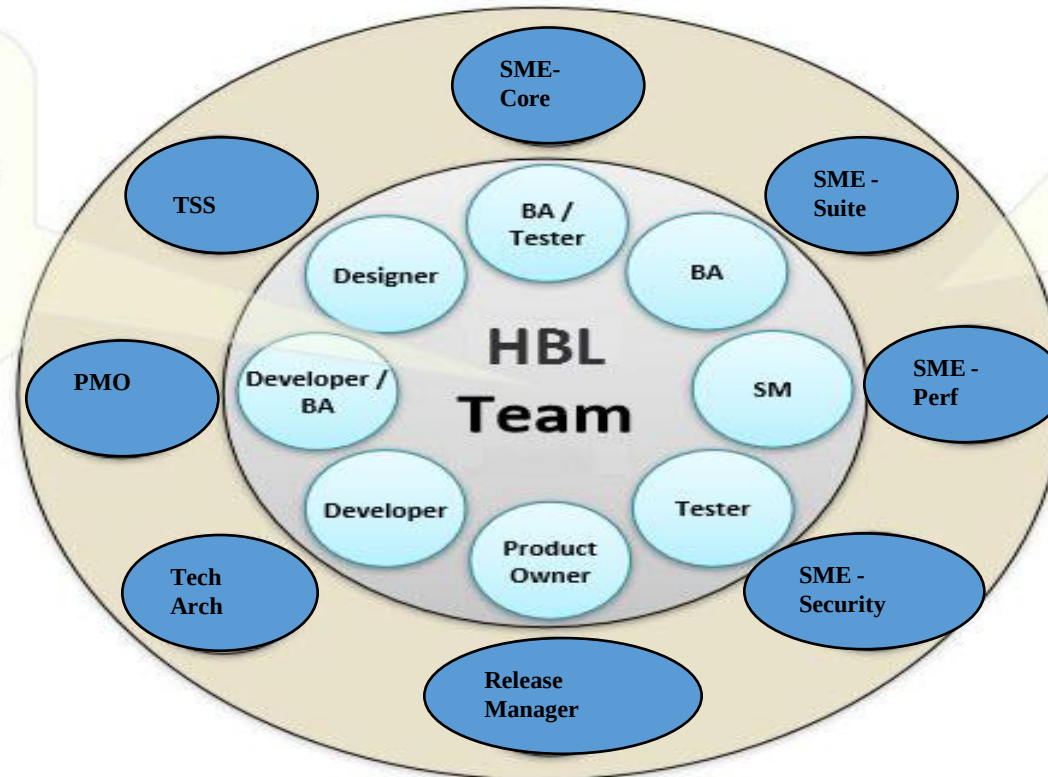


LEAP

Traditional Silos



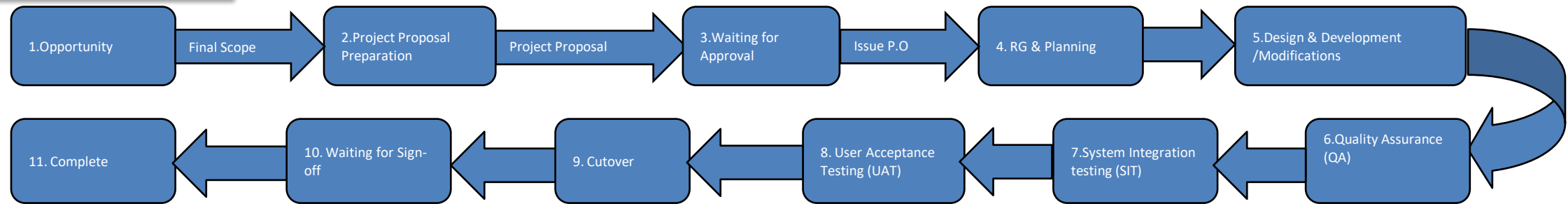
The HBL Project Team ideally consists of 5-9 dedicated members (7 +/- 2).



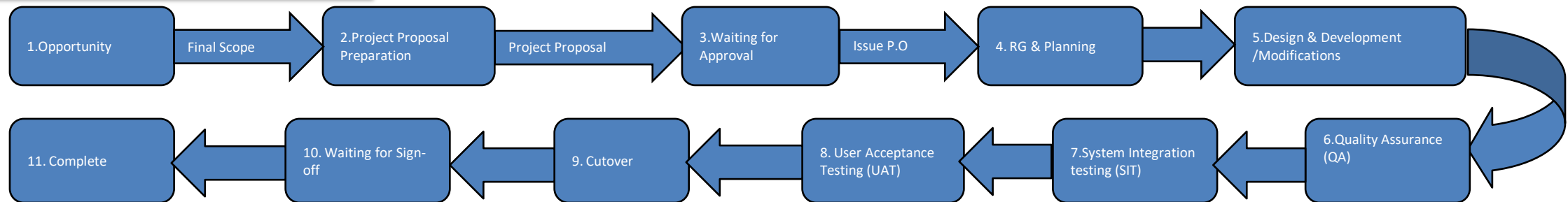
The **Extended Team** can contain many additional members, each playing an important role, but they are typically not dedicated to the effort.

Current Methodology (Cont)

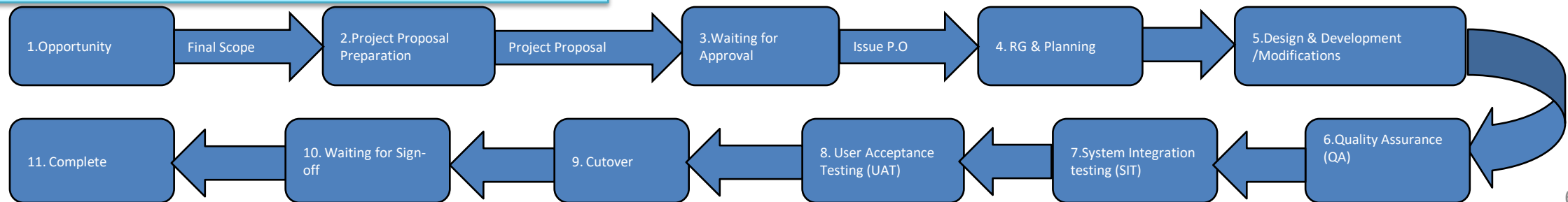
HL - UPOP:



HL - E- Commerce:



HL - Biometric Authentication

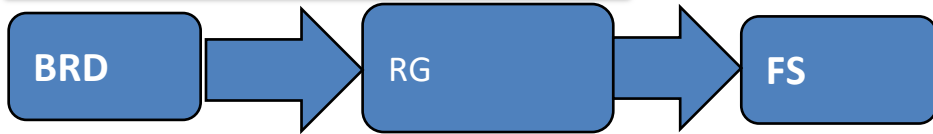


Agile Methodology for HBL

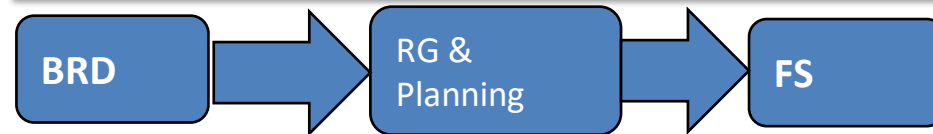
HBL - UPOP:



HBL - E- Commerce:



HBL – Biometric Authentication



High Level Scrum Process

